

ME5412 Robotics for Healthcare

Principles for Assistive Technologies

辅助技术原理.

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Principles for Assistive Technologies

- WHO definitions • 世界卫生组织 (WHO) 相关定义
- Definition of Assistive Technology 辅助技术的定义
- HAAT model HAAT模型 通用设计原则与以用户为
- Principles of Universal Design and user-centred design 中心的设计
- Assistive Technology characterization 辅助技术的特征说明
- Robotics as Assistive Technology 机器人作为辅助技术

残疾关键事实

Disability key facts

- An estimated **1.3 billion** people experience significant disability. This represents **16%** of the world's population, or 1 in 6 of us.
- Some persons with disabilities die up to 20 years earlier than those without disabilities.
- Persons with disabilities have twice the risk of developing conditions such as depression, asthma, diabetes, stroke, obesity or poor oral health.
- Persons with disabilities face many health inequities.
- Persons with disabilities find **inaccessible** and unaffordable transportation 15 times more difficult than for those without disabilities.
- Health inequities arise from unfair conditions faced by persons with disabilities, including **stigma, discrimination, poverty, exclusion from education and employment**, and **barriers** faced in the health system itself.
- 估计有13亿人存在显著的残疾状况，占全球人口的16%，即每6个人中就有1人。
- 部分残障人士的寿命相比无残疾人群最多可缩短20年。
- 残障人士罹患抑郁症、哮喘、糖尿病、中风、肥胖症或口腔疾病的风险高出一倍。
- 残障人群面临诸多健康不公问题。
- 残障人士遭遇交通设施无法通行、交通成本高昂的困难，概率是普通人群的15倍。
- 健康不公源于残障人士所面对的不公处境：社会污名、歧视、贫困、受教育与就业机会被剥夺，以及医疗系统本身存在的各类障碍。

<https://wecapable.com/types-of-disabilities-list/>

Increasing disability

残疾人数不断上升，原因：

Disability is increasing due to:

- longer life expectation, with increased prevalence of health conditions and diseases 人均预期寿命延长，各类健康问题与疾病的患病率随之上升
- increased risks of traumas, in car accidents and hazardous sports 交通事故、高危运动造成创伤的风险增加
- increased survival rates from traumatic injury 创伤性损伤后的生存存活率提高

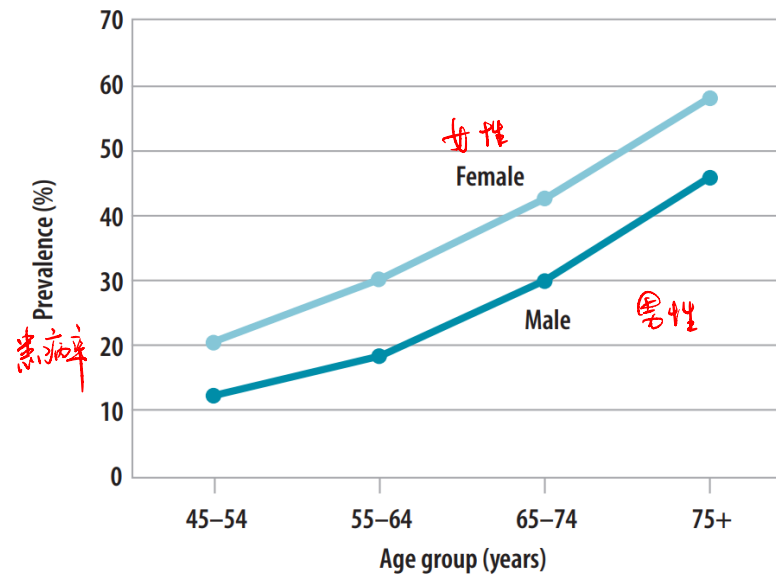
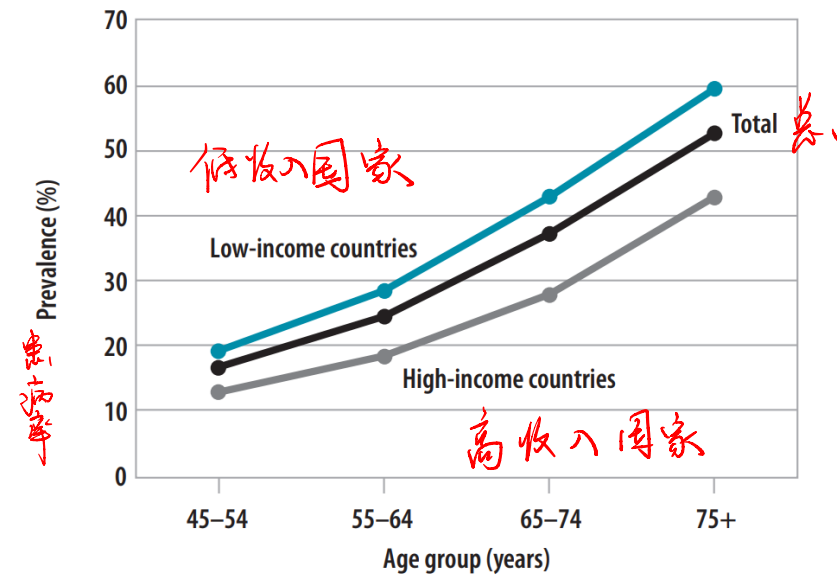
Increased need for personal assistance 个人照护援助需求增加

残疾与老龄化

Disability and ageing

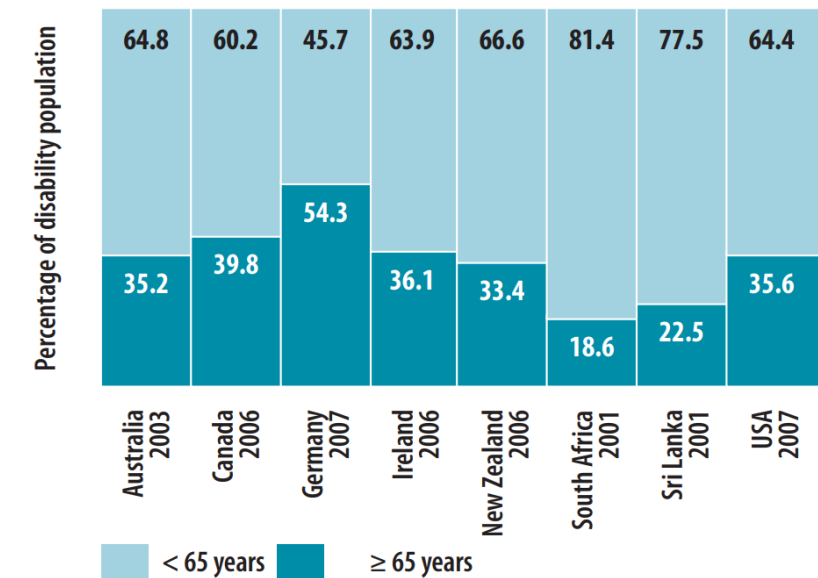
图2.2: 59个国家基于多维度功能水平统计, 按国家收入水平、性别划分的分年龄残疾患病率

Fig. 2.2. Age-specific disability prevalence, derived from multidomain functioning levels in 59 countries, by country income level and sex



不同国家残障人口的年龄分布占比

Fig. 2.3. Distribution of ages within disability populations



世界卫生组织国际分类体系

WHO international classifications

目标：为临床诊断、流行病学研究、临床评估以及康复项目提供统一标准术语

Objectives: to provide a uniform language for diagnostic practice, epidemiological studies, clinical assessment and rehabilitation programs

ICIDH
INTERNATIONAL CLASSIFICATION OF
IMPAIRMENTS, DISABILITY AND HANDICAP
1980

ICIDH 《国际损伤、残疾和障碍分类》 1980

ICIDH-2 《国际损伤、个人活动与社会参与分类》 1997

ICIDH-2
INTERNATIONAL CLASSIFICATION OF
IMPAIRMENTS, OF PERSONAL ACTIVITIES
AND OF SOCIAL PARTICIPATION
1997

ICF
INTERNATIONAL CLASSIFICATION
OF FUNCTIONING, DISABILITY
AND HEALTH
ICF 《国际功能、残疾和健康分类》 2001 2001

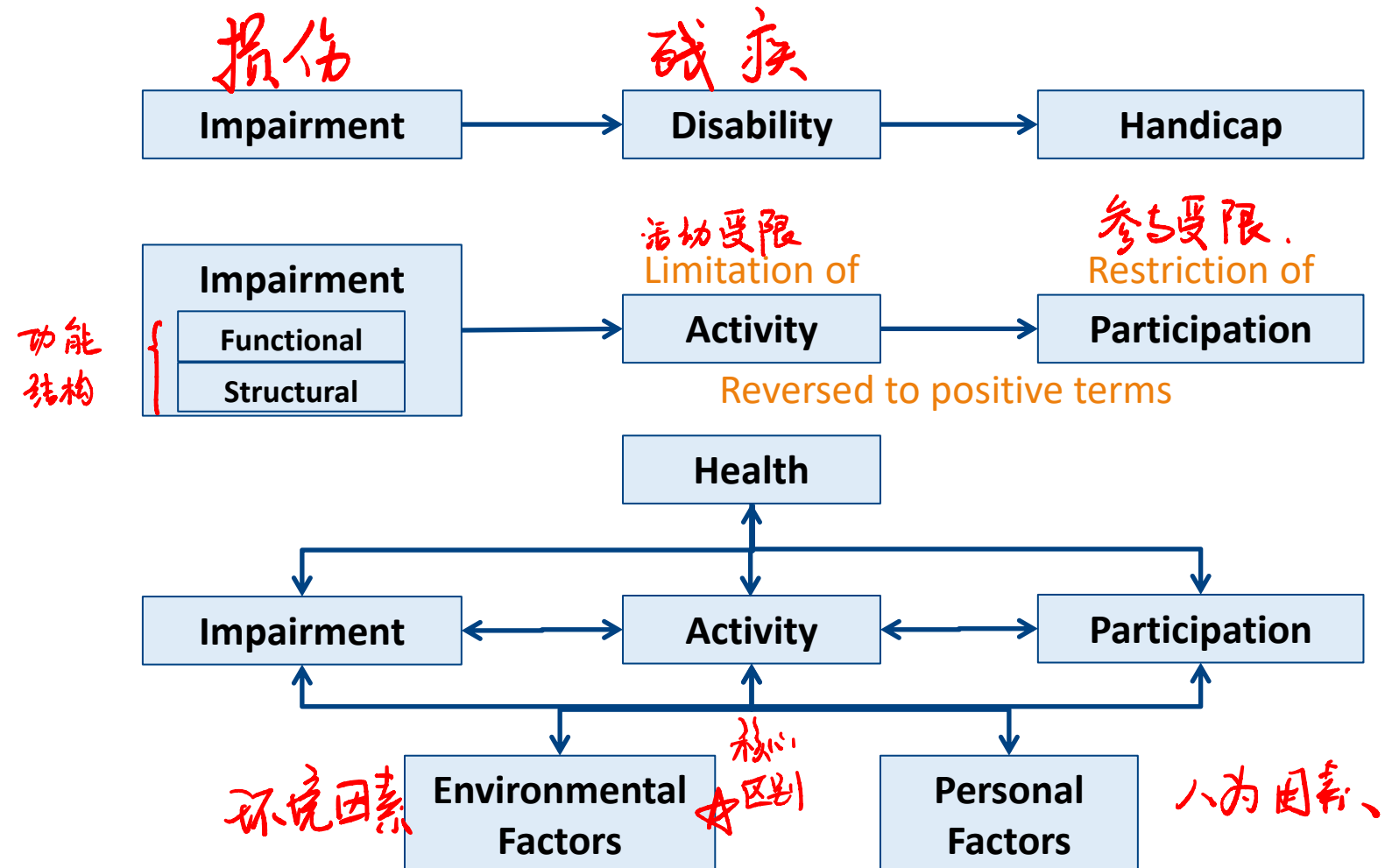
WHO international classifications

三套国际分类.

- ICIDH (1980)

- ICIDH2 (1997)

- IFC (2001)



WHO International Classification of Impairments, Disability and Handicap

ICIDH (1980)



疾病带来损伤：身体结构/外观异常，或是器官、系统功能故障

- A disease leads to an **impairment**: abnormality of a body structure or appearance or malfunction of an organ or system)
- An impairment leads to a **disability**: limitation to functional performance and activity of the individual 损伤带来残疾：个人执行功能、开展活动受到限制
- A disability leads to a **handicap**: disadvantage experienced by the individual as a result of the individual's disability and impairments

Focus on the impairment that prevents an individual from performing

残疾带来障碍：个体因为自身损伤与残疾，在社会中处于不利处境

该分类重点关注：造成个体行动受限的身体损伤本身。

WHO International Classification of Impairments, Personal Activities and Social Participation

ICIDH2 (1997)



- 损伤从功能、结构两个维度考量
 - Impairment is considered in its functional and structural aspects
 - Disability and handicap are replaced by positive terms, and expressed as limitations:
 - Activity limitation
 - Participation restriction
- 不再使用“残疾”“障碍”，改用正向概念表述，定义为两类受限：
- 活动受限
 - 参与受限

Focus on the barriers erected by society that hinder full participation of people with disabilities

关注重点：社会构建出的壁垒，阻碍残障人士充分参与社会生活

WHO International Classification of Functioning, Disability and Health

ICF (2001)

采用生物-心理-社会模型视角

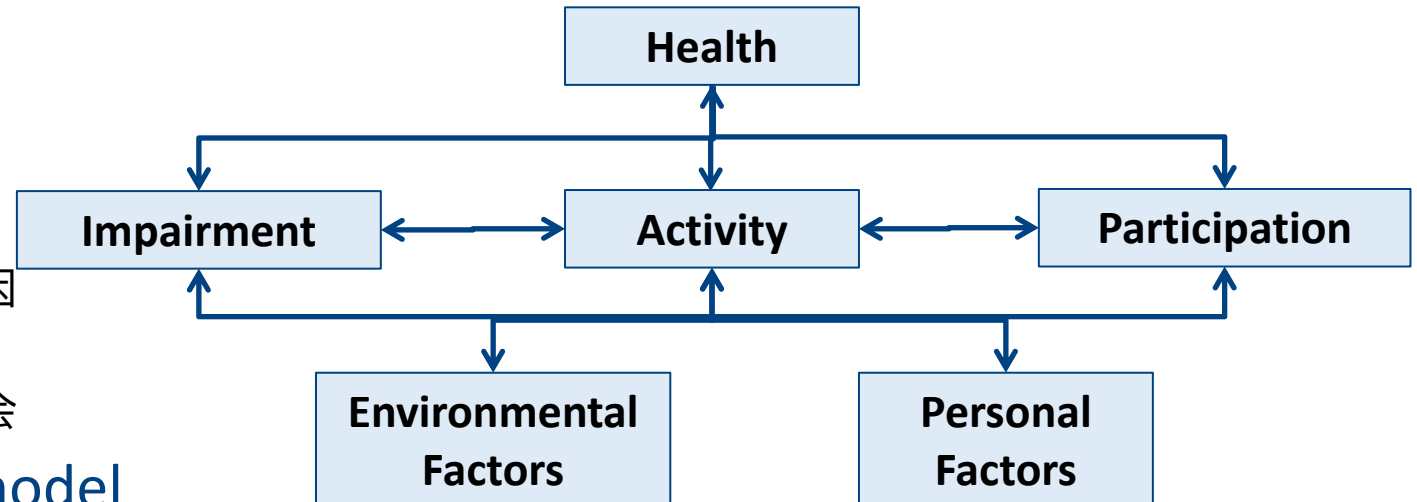
- 残疾涵盖损伤、活动受限、参与受限三个方面
- 残疾代表：个人健康状况，与环境、个人背景因素交互之后产生的负面结果
- 功能状态包含身体机能、身体结构、活动、社会参与

参与

Focus on a bio-psycho-social model

- **Disability** covers impairment, activity and participation
- Disability reflects the negative aspects of the interaction between the individual's health condition and contextual (environmental and personal) factors
- **Functioning** covers body functions, body structures, activities, participation
- Functioning denoted positive or neutral aspects of the interaction between the individual's health condition and contextual (environmental and personal) factors

功能状态代表：个人健康状况，与环境、个人背景因素交互之后中性或积极的一面



ICF in the WHO World Report on Disability

ICF 的 Report.

Box 1.1. New emphasis on environmental factors

The *International Classification of Functioning, Disability and Health* (ICF) (17) advanced the understanding and measurement of disability. It was developed through a long process involving academics, clinicians, and – importantly – persons with disabilities (18). The ICF emphasizes environmental factors in creating disability, which is the main difference between this new classification and the previous *International Classification of Impairments, Disabilities, and Handicaps* (ICIDH). In the ICF, problems with human functioning are categorized in three interconnected areas:

- **impairments** are problems in body function or alterations in body structure – for example, paralysis or blindness;
- **activity limitations** are difficulties in executing activities – for example, walking or eating;
- **participation restrictions** are problems with involvement in any area of life – for example, facing discrimination in employment or transportation.

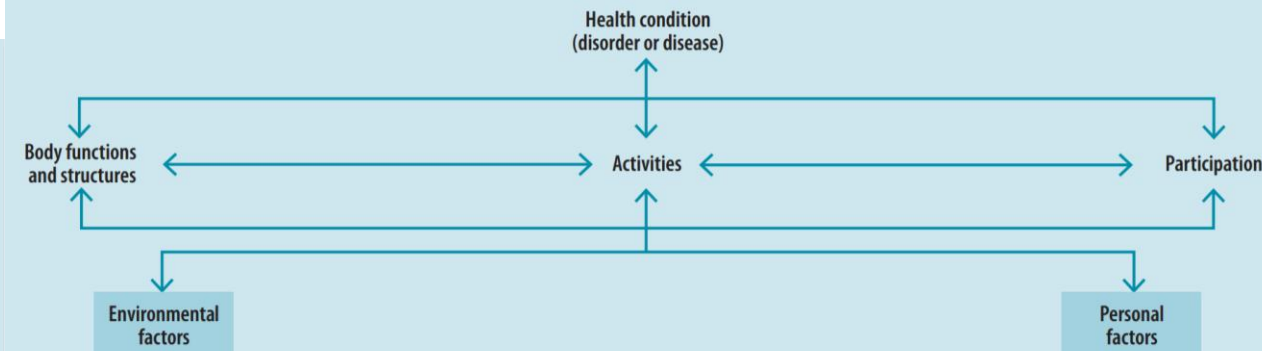
Disability refers to difficulties encountered in any or all three areas of functioning. The ICF can also be used to understand and measure the positive aspects of functioning such as body functions, activities, participation and environmental facilitation. The ICF adopts neutral language and does not distinguish between the type and cause of disability – for instance, between “physical” and “mental” health. “**Health conditions**” are diseases, injuries, and disorders, while “**impairments**” are specific decrements in body functions and structures, often identified as symptoms or signs of health conditions.

Disability arises from the interaction of health conditions with contextual factors – environmental and personal factors as shown in the figure below.

ICF将人体功能问题划分为互相联系的三大领域：

1. 损伤：身体功能出现问题，或是身体结构发生改变。例如瘫痪、失明。
2. 活动受限：执行具体活动存在困难。例如走路、吃饭。
3. 参与受限：在生活各个领域融入社会遇到阻碍。例如求职、乘坐公共交通遭受歧视。

Representation of the International Classification of Functioning, Disability and Health



The ICF contains a classification of **environmental factors** describing the world in which people with different levels of functioning must live and act. These factors can be either facilitators or barriers. Environmental factors include: products and technology; the natural and built environment; support and relationships; attitudes; and services, systems, and policies.

The ICF also recognizes **personal factors**, such as motivation and self-esteem, which can influence how much a person participates in society. However, these factors are not yet conceptualized or classified. It further distinguishes between a person's **capacities** to perform actions and the actual **performance** of those actions in real life, a subtle difference that helps illuminate the effect of environment and how performance might be improved by modifying the environment.

The ICF is universal because it covers all human functioning and treats disability as a continuum rather than categorizing people with disabilities as a separate group: disability is a matter of more or less, not yes or no. However, policy-making and service delivery might require thresholds to be set for impairment severity, activity limitations, or participation restriction.

It is useful for a range of purposes – research, surveillance, and reporting – related to describing and measuring health and disability, including: assessing individual functioning, goal setting, treatment, and monitoring; measuring outcomes and evaluating services; determining eligibility for welfare benefits; and developing health and disability surveys.

Which are true about disability?

0

→ 只是 Disability 的一个组成部分.

Disability is an impairment of body functions or structures

0%

Disability concerns impairment, activity limitations and participation restrictions ✓

0%

Disability is a health condition (disorder or disease)

错误, 而非残疾本身

0%

Disability restricts a person's participation in social life ✓

0%

Disability and handicap are synonymous

不是, handicap 是 Disability 下的一个体现.

0%

Disability limits a person's activities in daily life ✓

0%

常见援助与支持服务类型

Common types of assistance and support

- 社区支持与独立生活：协助自我照料、家务、行动移动、休闲活动、社区参与
- **Community support and independent living** – assistance with self-care, household care, mobility, leisure, and community participation;
- **residential support services** – independent housing and congregate living in group homes and institutional settings; • 喘息照护服务：为照护者、残障人士提供短期休整
- **respite services** – short-term breaks for caregivers and people with disabilities;
- **support in education or employment** – such as a classroom assistant for a child with a disability, or personal support in the workplace; 教育、就业支持：残障儿童课堂助教，职场个人辅助
- **communication support** – such as sign-language interpreters; 沟通支持：手语翻译等
- **community access** – including day care centres; 社区可及服务：日间照料中心
- **information and advice services** – including professional, peer support, advocacy, and supported decision-making; 信息咨询：专业咨询、同伴互助、维权倡导、辅助决策
- **assistance animals** – such as dogs trained to guide people with a visual impairment. 辅助动物：例如经过训练的导盲犬

UN Convention on the rights of persons with disabilities (CRPD)

第26条 能力建设与康复：

“.....采取适当措施（包括同伴支持），使残障人士能够实现并维持最大限度独立，充分发展身体、心理、社会、职业能力，完整融入并参与生活各个方面。

Article 26, Habilitation and Rehabilitation:

“... appropriate measures, including through peer support, to enable persons with disabilities to attain and maintain their maximum independence, full physical, mental, social and vocational ability, and full inclusion and participation in all aspects of life”.

- Services should be provided in the community, not in segregated settings 服务应当设置在社区内，而不是隔离式机构环境
- Person-centred services are preferable, so that individuals are involved in decisions about the support they receive and have maximum control over their lives.

优先采用以个人为中心的服务模式，让服务对象参与援助方案决策，最大限度掌握自己生活的主导权。

康复的若干定义

A few definitions around rehabilitation

世界卫生组织2023：康复通过优化人的功能状态，减轻残疾带来的影响，以此应对健康问题对日常生活造

- **Rehabilitation** addresses the impact of a health condition on a person's everyday life by optimizing their functioning and reducing their experience of disability (WHO 2023) 成的冲击。
- **Rehabilitation**: process aimed at enabling people with disability to reach and maintain their optimal physical, sensory, intellectual, psychological and social functional levels (WHO 2016) 世界卫生组织2016：康复是一套过程，帮助残障人群达到并维持最优的身体、感官、智力、心理以及社会功能水平。
- **Rehabilitation**: a set of measures that assist individuals who experience, or are likely to experience, disability to achieve and maintain optimal functioning in interaction with their environments (WHO Report on Disability, 2011) 《世界残疾报告》2011：康复是一系列干预措施，帮助当下或未来可能出现残疾的人，在和环境交互过程中实现、维持最优功能水平。

WHO World Report on Disability <https://www.who.int/publications/i/item/9789241564182>

World Health Organization (WHO) (2023) "Health topics: rehabilitation", <https://www.who.int/rehabilitation/en/>

C.J. Robinson, 1993. "Rehabilitation Engineering – an editorial", *IEEE Transactions on Rehabilitation Engineering*, Vol.1, pp.1-2.

J. Reswick, 1982. "What is a rehabilitation engineer?", in *Annual Review of Rehabilitation*, Vol.2, E.L. Pan, T.E. Backer, C.L. Vash (Eds.), Springer-Verlag, New York.

康复的四种实现策略

A few definitions around rehabilitation

康复弥补残障人士自身能力，和目标活动要求之间的差距，可以采用下面一种或多种策略：

- **Rehabilitation** aims at filling the gap between the capabilities of an individual with disabilities and the requirements of an activity the person wants to perform, through one or a combination of the following four strategies (EUSTAT 1999):
 1. 通过训练，提升使用者自身能力
 - Receive **training** to improve the individual's capabilities
 2. 改造环境，降低活动本身对人的能力要求
 - Change the **environment** to reduce the requirements of the activities
 3. 使用合适的辅助技术完成活动
 - Use adequate **assistive technology** to perform the activity
 4. 安排照护人员协助或者代为完成活动
 - Have a personal **helper** do or help with the activity
- **Rehabilitation** is the (re)integration of an individual with a disability into society (Robinson, 1993)

Robinson 1993：康复是帮助残障人士重新融入社会。

细分概念

A few definitions around rehabilitation

康复治疗：恢复、补偿功能损失，延缓身体各项机能衰退，覆盖生活全部领域（WHO世界残疾报告2011）。

康复工程（Reswick 1982）：运用科学技术手段减轻残障人士面临的障碍。

- **Rehabilitation Therapy** is concerned with restoring and compensating for the loss of functioning, and preventing or slowing deterioration in functioning in every area of a person's life (WHO Report on Disability, 2011).
- **Rehabilitation Engineering** is the application of science and technology to ameliorate the handicaps of individuals with disabilities (Reswick, 1982) 康复技术（辅助技术）：针对残障个体，挑选、设计、制作代
- **Rehabilitation Technology (or Assistive Technology)** is the selection, design, or manufacture of 偿/辅助设备。 augmentative or assistive devices that are appropriate for the individual with a disability
- **Assistive technology** enables and promotes inclusion and participation, especially of persons with disability, aging populations, and people with non-communicable diseases. The primary purpose of assistive products is to maintain or improve an individual's functioning and independence, thereby promoting their well-being. They enable people to live healthy, productive, independent and dignified lives, and to participate in education, the labour market and civic life (WHO Report on Disability, 2011).

辅助技术促进包容与社会参与，尤其面向残障人群、老年群体、非传染性疾病患者。辅助产品核心目标：维持、提升个人功能独立性，增进福祉；使人健康、高效、有尊严地独立生活，参与教育、就业、公共社会活动。（WHO世界残疾报告2011）

辅助技术 ☆

Assistive Technology (AT)

技术：用于辅助设备或者辅助相关服务

Technology* designed to be utilized in an assistive technology device or assistive technology service

- **Assistive technology device:** any item, piece of equipment, or product system, whether acquired commercially, modified, or customized, that is used to increase, maintain, or improve functional capabilities of individuals with disabilities.

United States Assistive Technology Act (1998, amended 2004)

- **Assistive technology device:** any product, instrument, equipment, or technology adapted or specially designed for improving the functioning of a disabled person.

辅助技术设备 (WHO 2001)：任何经过改造或者专门设计，改善残障人群功能状态的产品、器械、装备或技术。 WHO (2001)

- **Assistive technology service:** any service that directly assists an individual with disabilities in the selection, acquisition, or use of an assistive technology device.

United States Assistive Technology Act (1998, amended 2004)

*Technology: the practical application of knowledge especially in a particular area – not a machine or piece of equipment that is created by technology → technology device

普遍
含义

Common
meaning
课程关注
Our
focus



辅助设备示例

Examples of assistive devices

- 行动障碍人群：拐杖、假肢、矫形器、轮椅、三轮代步车
- 听力障碍人群：助听器、人工耳蜗
- 视力障碍人群：盲杖、放大镜、助视器具、有声读物、屏幕放大与阅读软件
- Crutches, prostheses, orthoses, wheelchairs, and tricycles for people with mobility impairments;
- hearing aids and cochlear implants for those with hearing impairments;
- white canes, magnifiers, ocular devices, talking books, and software for screen magnification and reading for people with visual impairments;
- communication boards and speech synthesizers for people with speech impairments; 言语障碍人群：沟通板、语音合成器
- devices such as day calendars with symbol pictures for people with cognitive impairment. 认知障碍人群：带图形符号的日程日历

Assistive technologies, when appropriate to the user and the user's environment, have been shown to be powerful tools to increase independence and improve participation.

适配用户与使用环境的辅助技术，能够有效提升独立性，促进社会参与。
辅助技术需要贴合用户需求；辅助技术并不能修复身体本身的损伤，目标是最大化活动能力与社会参与。

- Should respond to user's needs
- Do not improve impairment
- Maximise activity and participation



康复流程

Rehabilitation process

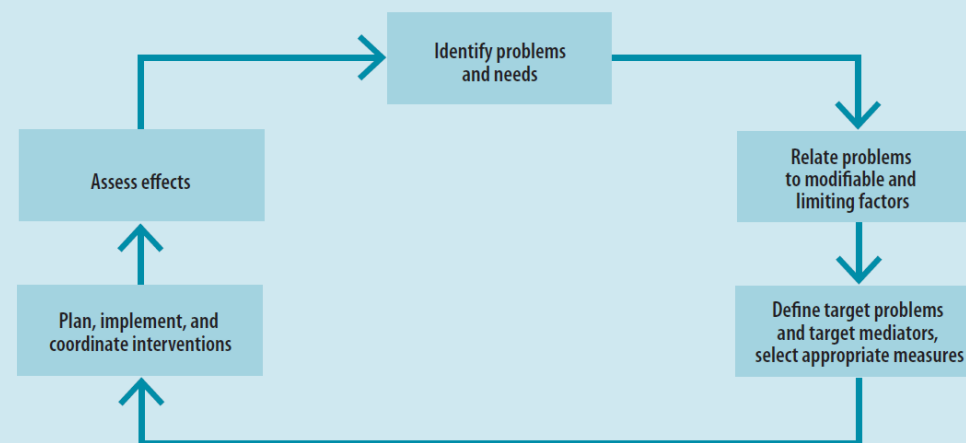
Box 4.1. What is rehabilitation?

This Report defines **rehabilitation** as “a set of measures that assist individuals who experience, or are likely to experience, disability to achieve and maintain optimal functioning in interaction with their environments”. A distinction is sometimes made between habilitation, which aims to help those who acquire disabilities congenitally or early in life to develop maximal functioning; and rehabilitation, where those who have experienced a loss in function are assisted to regain maximal functioning (2). In this chapter the term “rehabilitation” covers both types of intervention. Although the concept of rehabilitation is broad, not everything to do with disability can be included in the term. Rehabilitation targets improvements in individual functioning – say, by improving a person’s ability to eat and drink independently. Rehabilitation also includes making changes to the individual’s environment – for example, by installing a toilet handrail. But barrier removal initiatives at societal level, such as fitting a ramp to a public building, are not considered rehabilitation in this Report.

Rehabilitation reduces the impact of a broad range of health conditions. Typically rehabilitation occurs for a specific period of time, but can involve single or multiple interventions delivered by an individual or a team of rehabilitation workers, and can be needed from the acute or initial phase immediately following recognition of a health condition through to post-acute and maintenance phases.

Rehabilitation involves identification of a person’s problems and needs, relating the problems to relevant factors of the person and the environment, defining rehabilitation goals, planning and implementing the measures, and assessing the effects (see figure below). Educating people with disabilities is essential for developing knowledge and skills for self-help, care, management, and decision-making. People with disabilities and their families experience better health and functioning when they are partners in rehabilitation (3–9).

The rehabilitation process



Source: A modified version of the Rehabilitation Cycle from (10).

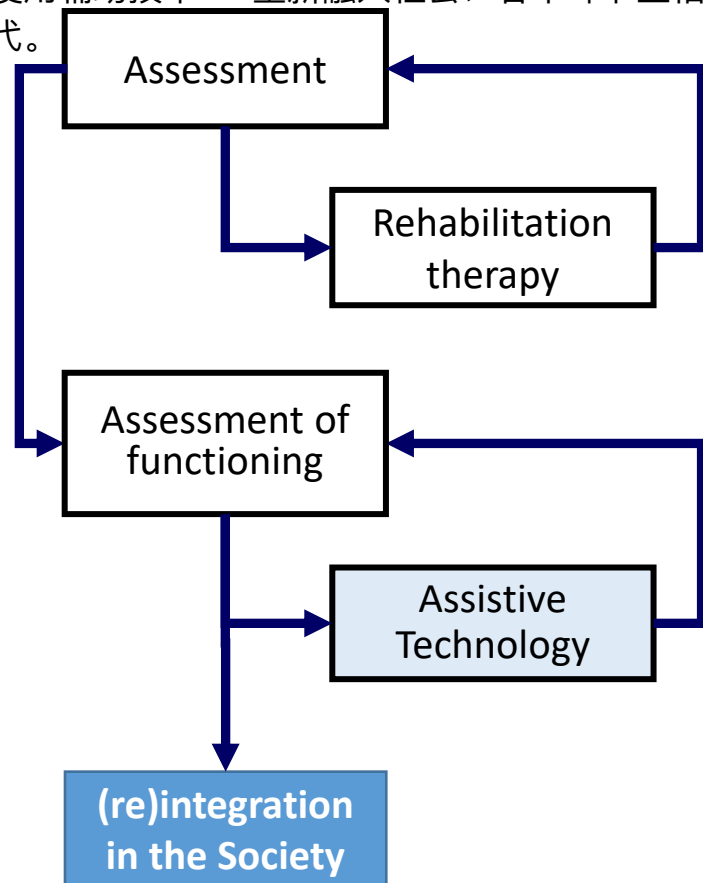
Rehabilitation – provided along a continuum of care ranging from hospital care to rehabilitation in the community (12) – can improve health outcomes, reduce costs by shortening hospital stays (15–17), reduce disability, and improve quality of life (18–21). Rehabilitation need not be expensive.

Rehabilitation is cross-sectoral and may be carried out by health professionals in conjunction with specialists in education, employment, social welfare, and other fields. In resource-poor contexts it may involve non-specialist workers – for example, community-based rehabilitation workers in addition to family, friends, and community groups.

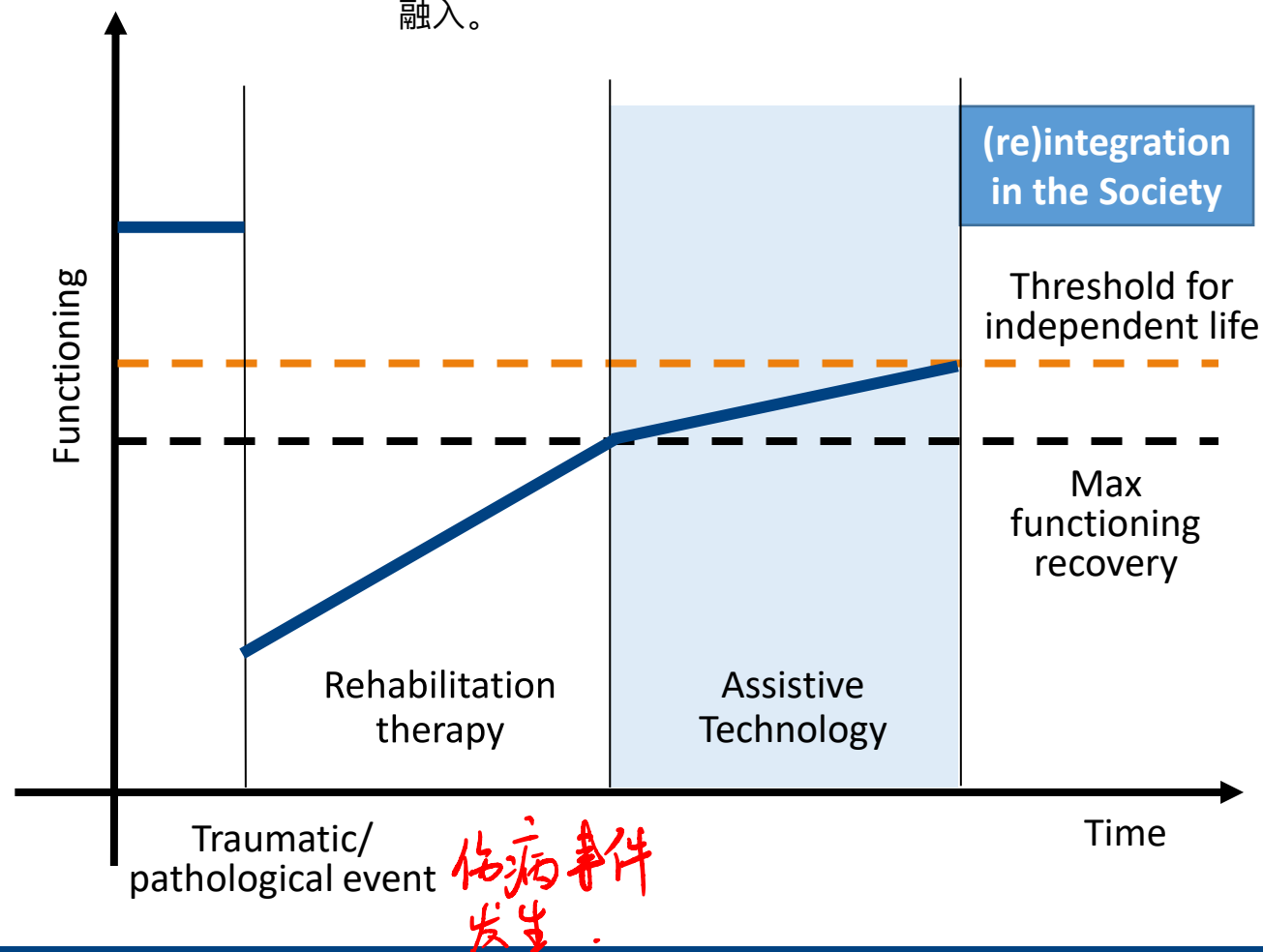
Rehabilitation that begins early produces better functional outcomes for almost all health conditions associated with disability (18–30). The effectiveness of early intervention is particularly marked for children with, or at risk of, developmental delays (27, 28, 31, 32), and has been proven to increase educational and developmental gains (4, 27).

Rehabilitation process

1) 康复闭环流程图：评估 → 康复治疗 → 功能评估 → 使用辅助技术 → 重新融入社会；各个环节互相反馈迭代。



2) 时间-功能曲线图：伤病事件发生，先做康复治疗提升功能，再借助辅助技术，使功能达到独立生活阈值，完成社会再融入。



辅助技术 ISO 标准分类

ISO Classification of Assistive Technology

The screenshot shows the ISO Standards catalogue website. The top navigation bar includes links for Standards, About us, News, Taking part, and Store, along with a search bar and a shopping cart icon. The main content area is titled '← Standards catalogue' and 'ICS'. It states that ISO has developed over 24876 International Standards and all are included in the ISO Standards catalogue. There is a section for 'Draft standards for public comment' which mentions that standards at the enquiry stage are open for comments and provides a link to a 'list of standards' and contact information for a 'national member'. Below this, there are two tabs: 'Browse by ICS' (selected) and 'Browse by TC'. A search bar labeled 'Search in the list' is present. A table lists various ICS fields, with '11 Health care technology' highlighted by an orange box.

ICS ↑	Field
01	Generalities. Terminology. Standardization. Documentation
03	Services. Company organization, management and quality. Administration. Transport. Sociology
07	Natural and applied sciences
11	Health care technology
13	Environment. Health protection. Safety
17	Metrology and measurement. Physical phenomena

ISO 9999:2022

ISO Classification of Assistive Technology

← ICS

11 

Health care technology

ISO 9999:2022
ICS 11 医疗保健技术

ICS ↑	Field
11.020	Medical sciences and health care facilities in general → Forensic science, see 07.140
11.040	Medical equipment
11.060	Dentistry
11.080	Sterilization and disinfection
11.100	Laboratory medicine
11.120	Pharmaceutics
11.140	Hospital equipment <i>Including hospital beds, surgical tables, medical garments, medical gloves, containers for sharp disposal, etc.</i>
11.160	First aid <i>Including kits, equipment, facilities and medical transport for first aid</i>
11.180	Aids for disabled or handicapped persons <i>Including aids for elderly people</i>

ISO 9999:2022
11.180

ISO Classification of Assistive Technology


Standards
About us
News
Taking part
Store



← ICS ← 11

11.180

残障人士、老年人辅助器具

Aids for disabled or handicapped persons

Including aids for elderly people

ICS ↑

Field

11.180.01

Aids for disabled and handicapped persons in general

11.180.10

Aids and adaptation for moving

Including wheelchairs, walking sticks and lifting platforms

11.180.15

Aids for deaf and hearing impaired people

11.180.20

Aids for incontinence and ostomy

11.180.30

Aids for blind or partially sighted people

Including Braille

→ Glasses, contact lenses and their cleaning products, see 11.040.70

11.180.40

Aids for drinking and eating

11.180.99

Other standards related to aids for disabled and handicapped people

• 11.180.01

残障辅助器具通用

• 11.180.10

移动辅助器具：轮椅、

• 11.180.15

听障辅助器具

• 11.180.20

失禁、造口护理辅助器

• 11.180.30

视障辅助器具（盲文等

• 11.180.40

饮食饮水辅助器具

• 11.180.99

其他残障辅助相关标准

ISO 9999:2022
11.180.xx

赋能环境

Enabling environments

1. 无障碍（可及性 Accessibility）：通俗含义：抵达、理解、接触事物的能力；法规标准层面指法律强制满足的要求。
2. 通用设计（Universal design）：一套设计流程，面对人群能力多样性，设计环境、产品、系统，提升可用性、安全性、健康水平、社会参与。

Box 6.1. Definitions and concepts

Accessibility – in common language, the ability to reach, understand, or approach something or someone. In laws and standards on accessibility, it refers to what the law requires for compliance.

Universal design – a process that increases usability, safety, health, and social participation, through design and operation of environments, products, and systems in response to the diversity of people and abilities (1).

Usability, though, is not the only goal of universal design, and “adaption and specialized design” are a part of providing customization and choice, which may be essential for addressing diversity. Other overlapping terms for the same general concept are “design for all” and “inclusive design”.

Standard – a level of quality accepted as a norm. Sometimes standards are codified in documents such as “guidelines” or “regulations”, both with specific definitions, with different legal implications in different legal systems. An example is Part M of the Building Regulations in the United Kingdom of Great Britain and Northern Ireland. Standards can be voluntary or compulsory.

Public accommodations – buildings open to and provided for the public, whether publicly owned (such as courts, hospitals, and schools) or privately owned (such as shops, restaurants, and sports stadia) as well as public roads.

Transportation – vehicles, stations, public transportation systems, infrastructure, and pedestrian environments.

Communication – “includes languages, text displays, Braille, tactile communication, large print, and accessible multimedia as well as written, audio, plain-language, human-reader and augmentative and alternative modes, means, and formats of communication, including accessible information and communication technology” (2). These formats, modes, and means of communication may be physical, but are increasingly electronic.



辅助技术 (AT) 能影响.

Impact of assistive technology

Box 4.3. Money well spent: The effectiveness and value of housing adaptations

Public spending on housing adaptations for people with difficulties in functioning in the United Kingdom of Great Britain and Northern Ireland amounted to more than £220 million in 1995, and both the number of demands and unit costs are growing. A 2000 research study examined the effectiveness of adaptations in England and Wales, using interviews with recipients of major adaptations, postal questionnaires returned by recipients of minor adaptations, administrative records, and the views of visiting professionals. The main measure of “effectiveness” was the degree to which the problems experienced by the respondent before adaptation were overcome by the adaptation, without causing new problems. The study found that:

- Minor adaptations (ramps, over-bath showers, and door entry systems, for example) – most costing less than £500 – produced a range of lasting, positive consequences for virtually all recipients: 62% of respondents suggested they felt safer from the risk of accident, and 77% perceived a positive effect on their health.
- Major adaptations (bathroom conversions, extensions, lifts, for example) in most cases had transformed people’s lives. Before adaptations, people used words like “prisoner”, “degraded”, and “afraid” to describe their situations; following adaptations, they spoke of themselves as “independent”, “useful”, and “confident”.
- Where major adaptations failed, it was typically because of weaknesses in the original specification. Adaptations for children sometimes failed to allow for the child’s growth, for example. In other cases, policies intended to save money resulted in major waste. Examples included extensions that were too small or too cold to use, and cheap but ineffective substitutes for proper bathing facilities.
- The evidence from recipients suggests that successful adaptations keep people out of hospitals, reduce strain on carers, and promote social inclusion.
- Benefits were most pronounced where careful consultation with users took place, where the needs of the whole family had been considered, and where the integrity of the home had been respected.

Adaptations appear to be a highly effective use of public resources, justifying investment in health and rehabilitation resources. Further research is needed in diverse contexts and settings.

Assistive technology and accessible environments

- Improve the quality of life of citizens with disability
- Increase the productivity of citizens with disability
- Reduce the assistance costs

辅助技术和无障碍环境的价值:

- 改善残障公民生活质量
- 提升残障公民劳动生产力
- 降低照护援助成本

辅助技术相关模型

Assistive technology models

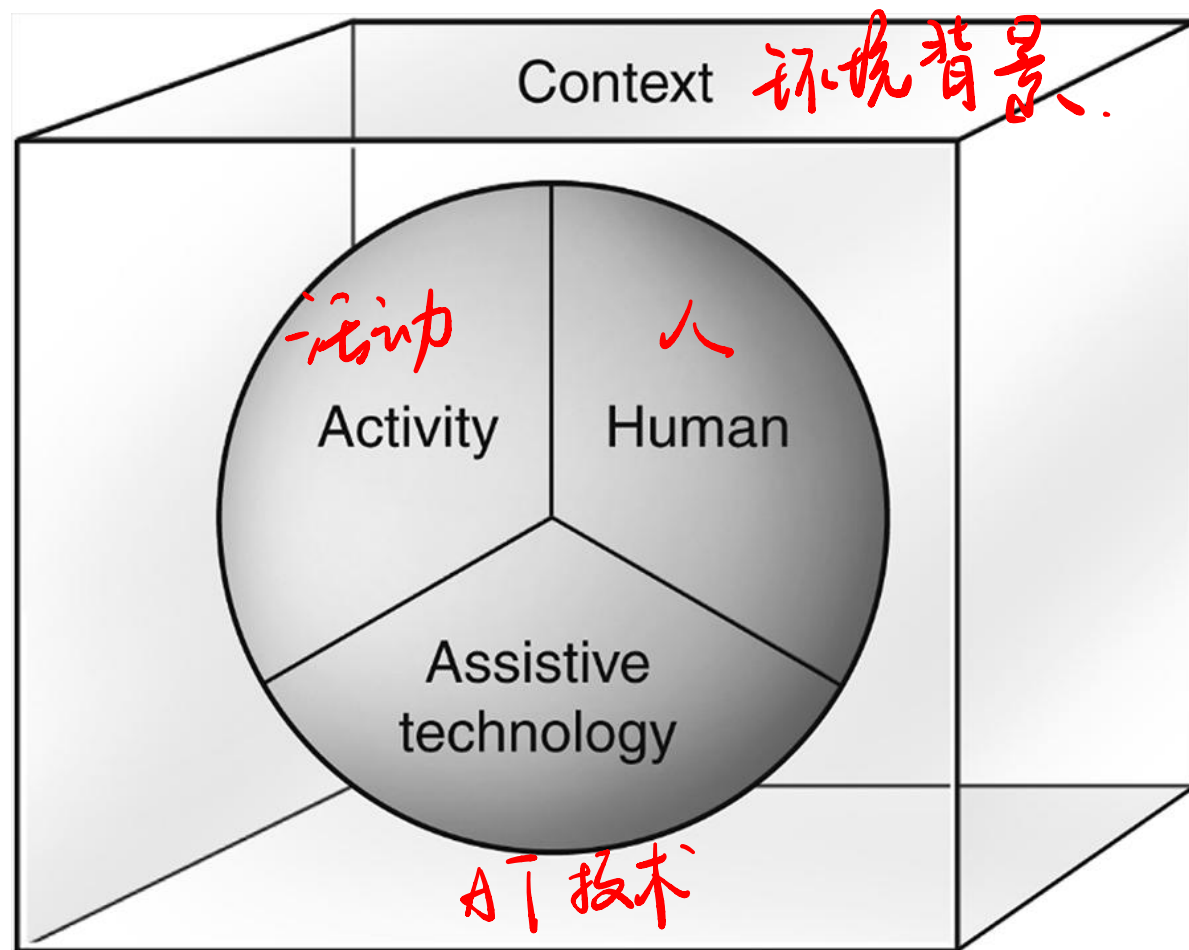
- HEART – Horizontal European Activities in Rehabilitation Technology (1994)
- HAAT – Human-Activity-AssistiveTechnology (1995)
- Enders model (1999)
- Dynamic circles of human activities (2006)

HEART (欧洲康复技术横向项目, 1994)

- HAAT 人-活动-辅助技术模型 (1995, Enders模型1999)
- 人类活动动态圆环模型 (2006)

HAAT 模型：人-活动-辅助技术

HAAT Model: Human-Activity-Assistive Technology



人是模型的核心：存在开展某一项活动的需求或意愿

任何活动都发生在特定背景环境中

- The **human** component is the core, with need/wish to perform an **activity**
- Any activity is done in a **context**
- The combination of activity and context outlines the human's required **skills**
活动+环境背景共同决定人需要具备哪些能力
- If human's residual skills are not enough for the activity in the context, **assistive technology (AT)** is required
如果人自身残余能力不足以在该环境完成这项活动，就需要辅助技术
- Since each human's skills are different, any AT device is unique
每个人能力各不相同，因此辅助设备都应当因人而异定制。

活动 (HAAT) Activity

核心要素

1. 定义辅助设备要达成的目标
2. 是人自身能力在功能层面的体现
3. 属于日常生活，可后天学习；受社会、文化环境背景影响

Fundamental element

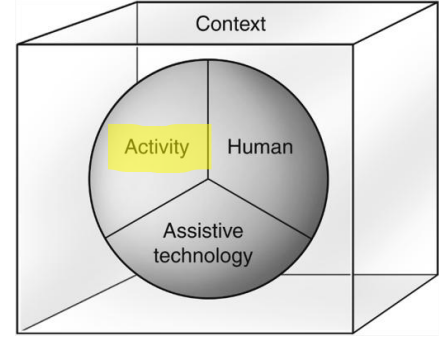
- It defines the objective of the assistive device
- It is the functional expression of the human's skills
- It is part of daily life, it can be learned, it is influenced by social and cultural factors related to the context

活动分类

1. 日常生活活动ADL：穿衣、个人卫生、进食、沟通
2. 工作生产类活动：家务管理、教育学习、职业工作、照料他人
3. 休闲娱乐活动

Activity classification

- Activities of daily life (ADLs):
 - Dressing, hygiene, eating, communication
- Work and productive activities:
 - Home management, educational, vocational, care-of-others activities
- Leisure activities

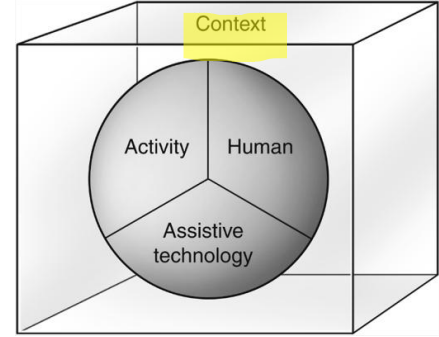


环境背景 (HAAT)

Context

环境背景四大属性：

1. 物理环境：自然、建筑环境要素，促进或者阻碍参与、辅助设备使用；噪声、光照、温度等物理参数。
2. 社会环境：环境中的其他人，直接或间接影响社会参与和辅助设备使用。
3. 文化环境：群体共有的信念、习俗、价值观，变化速度慢于社会行为。
4. 制度环境：法律法规；政策与经费保障。



Context description consists of 4 main attributes:

1. **Physical context:** elements of natural and built environments that support or hinder participation and AT use; it includes physical parameters like noise, light, temperature
2. **Social context:** individuals in the environment who affect active participation and AT use, directly or indirectly
3. **Cultural context:** systems of shared meanings that include beliefs, rituals, and values that are broadly held and do not change as quickly as socially held attributes and practices
4. **Institutional context:**
 1. Legislation and related regulations
 2. Policies and funding

Assistive Technology (AT)

任何辅助技术设备由4个核心部件组成：

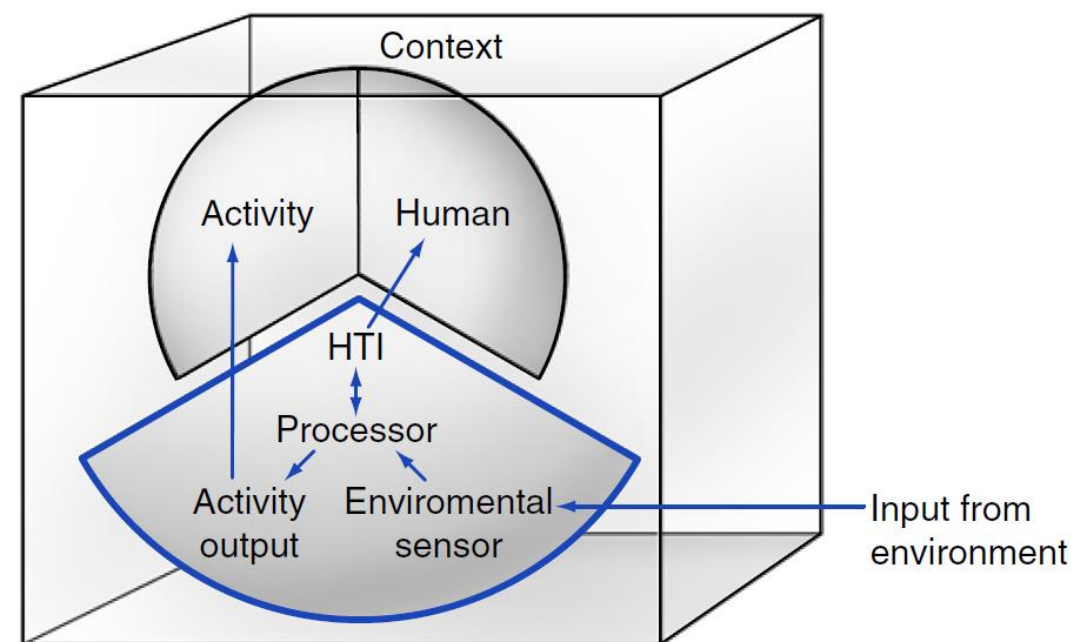
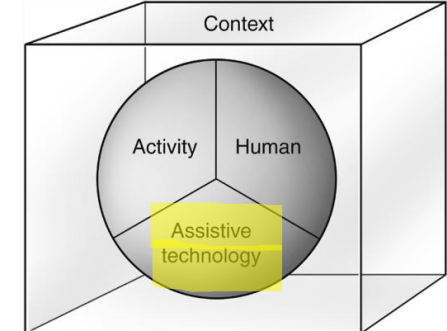
1. 人-辅助设备交互接口 HTI：可以物理层面或者认知层面，双向交互。
2. 活动输出模块：实现实际功能表现。

Any AT device consists of 4 main components:

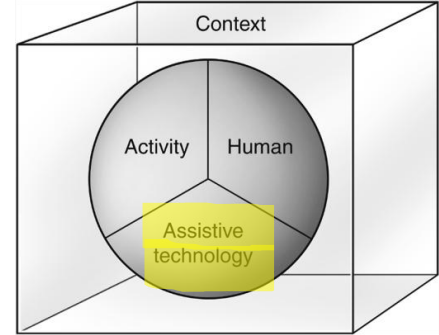
1. **Human-AT interface (HTI)**: it can be physical or cognitive, it is bi-directional
2. **Activity output**: it contributes to the functional performance
3. **Processor**: it translates physical (forces) and cognitive (information) inputs into signals used to control the activity output
4. **Environmental sensor**: it detects information on the context and transfers them to the processing unit and to the human, through the human-AT interface

3. 处理器：把物理输入（力）、认知输入（信息）转换成控制活动输出的信号。

4. 环境传感器：采集环境信息，传递给处理器，并通过人机接口反馈给使用者。



Assistive Technology (AT)



辅助技术赋能并促进包容参与，尤其面向残障人群、老年人、非传染性疾病患者。辅助产品首要目的：**维持、提升个人功能独立性，增进福祉**；让人过上健康、高效、独立、有尊严的生活，参与教育、劳动力市场与公共事务。

- Assistive technology **enables and promotes inclusion and participation**, especially of persons with disability, aging populations, and people with non-communicable diseases. The primary purpose of assistive products is **to maintain or improve an individual's functioning and independence**, thereby promoting their well-being. They enable people to live healthy, productive, independent and dignified lives, and to participate in education, the labour market and civic life.
- WHO estimates that today **2.5 billion people** need one or more assistive products such as wheelchairs, hearing aids, or apps that support communication and cognition. With a global ageing population and a rise in noncommunicable diseases, this number will rise beyond 3.5 billion by 2050, with many older people needing two or more products as they age.
- Examples of assistive products include hearing aids, wheelchairs, spectacles, prostheses and devices that support memory, among many others. While supporting **independence and well-being**, these products can also help to **prevent or reduce the effects of secondary health conditions**, such as lower limb amputation in people with diabetes. They can also **reduce the need and impact on carers** and mitigate the need for formal health and support services. Moreover, access to appropriate assistive products can have a tremendous impact on **community development and economic growth**.

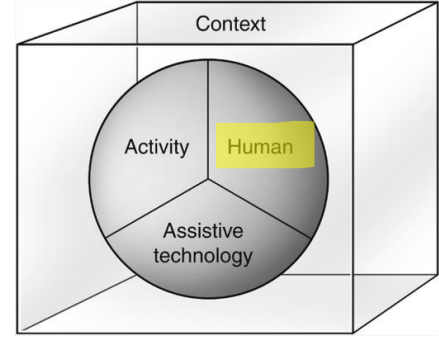
世界卫生组织估算：当前25亿人需要至少一类辅助产品，例如轮椅、助听器、沟通认知辅助软件。伴随全球老龄化、非传染性疾病增加，2050年该数字将超过35亿；很多老年人会同时需要2种以上辅助产品。

辅助产品举例：助听器、轮椅、眼镜、假肢、记忆辅助设备等等。

辅助技术维持独立与健康，还可以预防、减轻继发健康问题（例如糖尿病造成下肢截肢）；减轻照护者负担，减少正式照护服务需求；合适的辅助产品对社区发展、经济增长意义重大。

人 (HAAT)

Human component

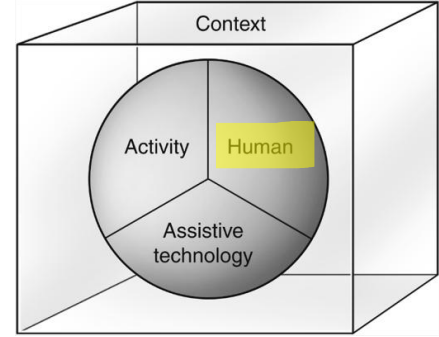


- The person with disabilities, with residual physical and cognitive abilities and emotional states
- He/she selects the activity to perform and the context
- Body functions and structures of ICF are included in the HAAT human component

- 残障使用者，具备残余生理、认知能力，带有情绪状态；由使用者选择想要开展的活动、所处环境。
- ICF分类中的身体功能、身体结构概念全部纳入HAAT模型的「人」组件。

“人”的能力与技巧。

Human abilities and skills



- **Ability** is a person's characteristic, visible in task accomplishment
- **Skill** is a specific expertise level acquired with practice
- **Residual abilities** can be assessed, while it is difficult to predict the skills a person can reach

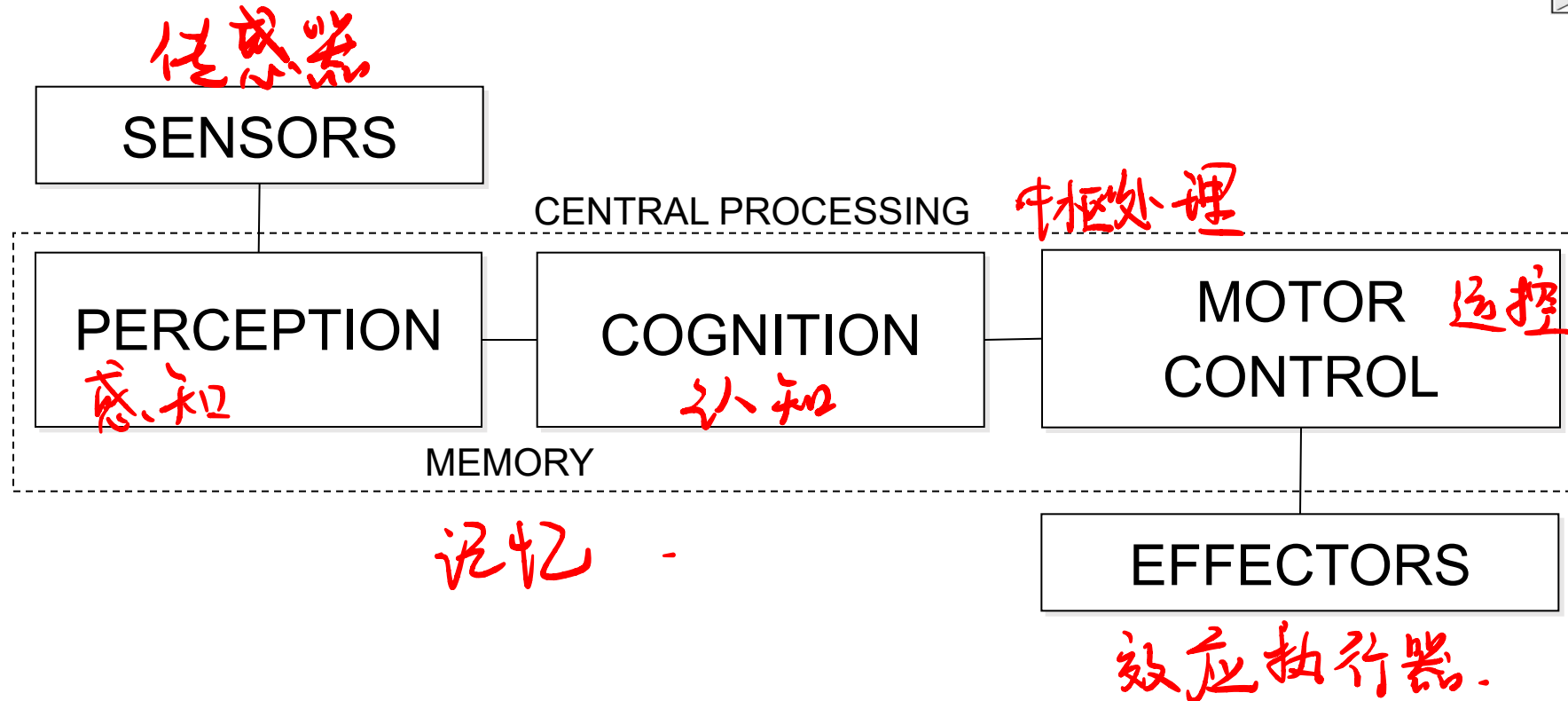
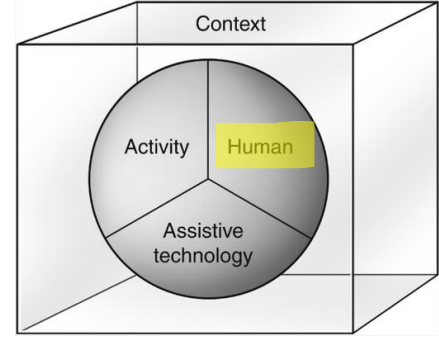
能力Ability: 人的内在特质, 体现在任务完成结果。

技能Skill: 通过练习获得的特定熟练水平。

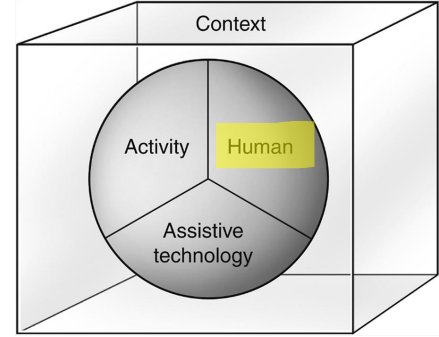
残余能力可以评估; 但人未来可以达到多高技能水平很难预判。

人的功能组件模型

A model of the human component



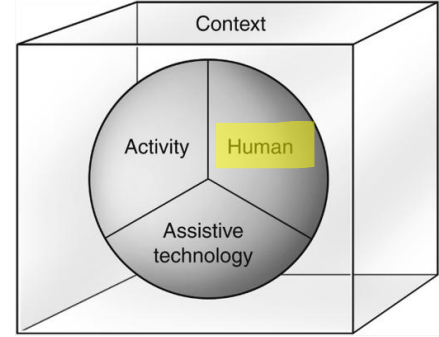
Sensors



- Inputs from the environment
- Possible limitations:
 - Sensitiveness (minimum detectable threshold)
 - Magnitude (detectable variation in size/magnitude)

1. 传感器：接收来自环境输入；
局限可以是感知灵敏度、可分辨信号幅度。

Effectors



- Neural, muscular and skeletal elements of the human body that provide movement or motor output
- They operate under control from central processing and in response to sensor inputs
- Possible limitations:
 - Impairment of one or a combination of elements

2. 效应执行器Effectors: 人体神经、肌肉、骨骼, 实现运动输出;
接收中枢处理与传感器信号;
缺陷来自身体部件损伤。

中枢处理

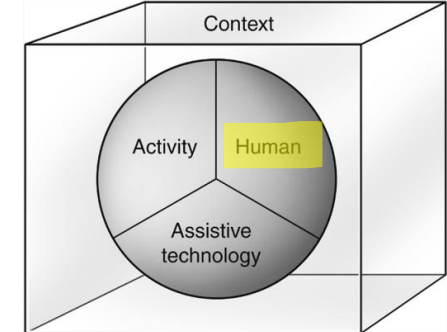
Central processing

包括：感知、认知、神经运动控制、心理因素

- 感知：结合记忆中的信息，对传感器接收的数据进行解读与理解
- 认知：注意力、记忆、问题求解、决策、学习、语言及其他功能

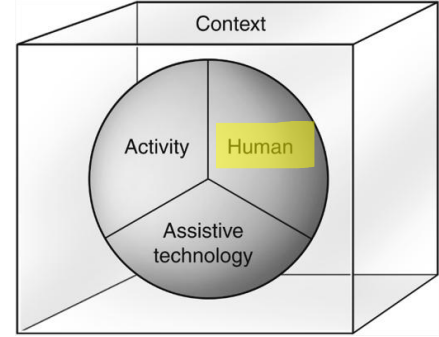
- Perception, cognition, neuro-motor control, psychological factors
- Perception: interpretation and understanding of data received from sensors, with use of information contained in the memory
- Cognition: attention, memory, problem solving, decision, learning, language and other functions
- Motor control: integration of sensory, motor and cognitive components in a motor action executed by the effectors
 - Motor planning: movements executed for accomplishing a task
 - Motor learning: acquisition of the skills for executing a task autonomously through practice
- Psycho-social functions: identity, self-protection and motivation

社会心理功能：自我认同、自我保护与动机



运动控制：在效应器执行动作时，整合感知、运动与认知模块

The human component in AT design



辅助技术设计时，设计师需要掌握使用者三点信息

For designing AT, the designer needs to know:

1. What the human can do (abilities and skills)
2. What the human cannot do (limitations)
3. What the human wants to do (motivations)
 - Motivations are directly connected to the human's objectives and needs and to how much an AT device responds to them
 - Abilities and limitations has to be considered with respect to the HAAT model areas

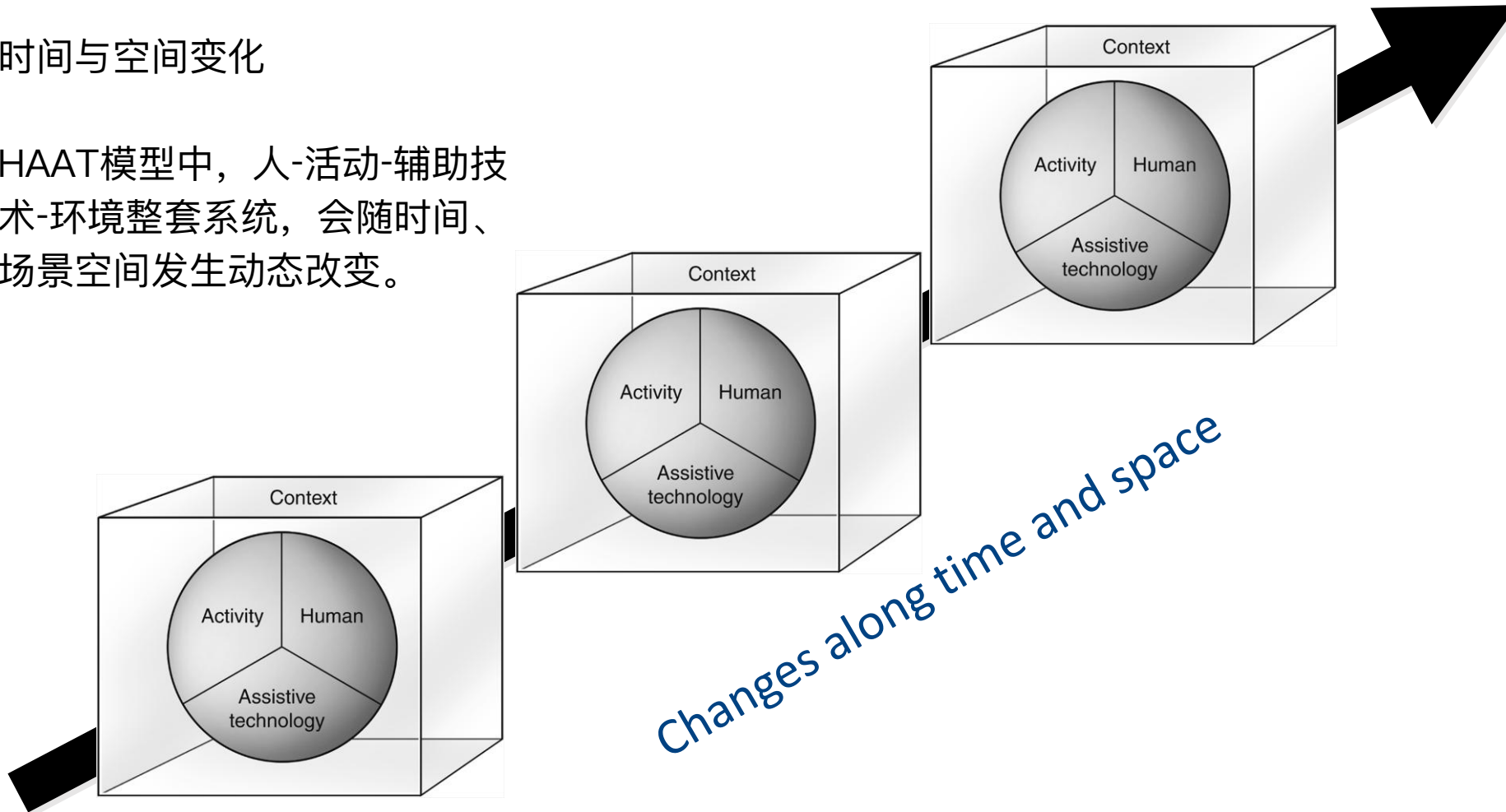
1. 人可以做到什么（能力与技能）
2. 人做不到什么（局限缺陷）
3. 人想要做什么（动机诉求）

动机直接关联个人目标需求，决定辅助设备多大程度满足使用者。
评估能力和局限时，需要结合HAAT模型全部维度。

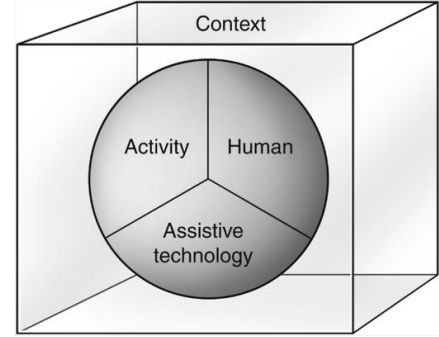
Time and space effects

时间与空间变化

HAAT模型中，人-活动-辅助技术-环境整套系统，会随时间、场景空间发生动态改变。



Application of the HAAT model

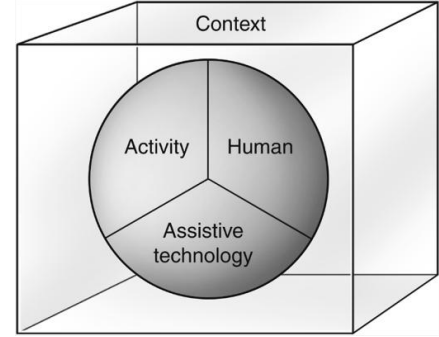


Four primary applications:

1. Product research and development
 2. Product usability studies
 3. Clinical assessment
 4. Outcome evaluation
- HAAT模型四大应用场景
1. 产品研发
 2. 产品可用性研究
 3. 临床评估
 4. 使用效果评价

1) 产品研发案例：带坐姿支撑的儿童坐便系统

Product research and development

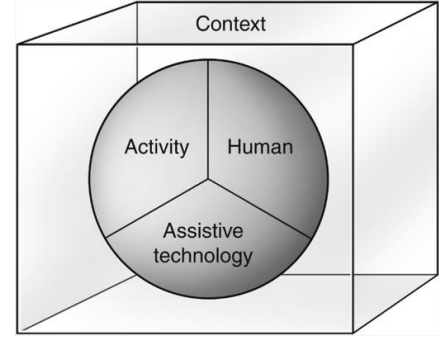


Example: toileting system for children with seating needs

1. Parents of young children providing their opinion on toileting needs, including how they needed to be able to use a toileting product (e.g., it needed to be easily removed from a toilet so others in the household could use the toilet).
2. These opinions were then translated into design features that led to the development of an initial prototype and an evaluation instrument that could be used to critique the prototype.
3. The prototype, along with other currently available products, was then evaluated by parents using the criteria that had been developed in the initial phase.
4. Concurrently, the children who might benefit from the device were seated in it so that an experienced physiotherapist could evaluate the positioning and the children could indicate their opinion of the device.
5. The final stage of the development of the product, before the technology was transferred, involved use of the device in the home over an extended timeframe so the parents and children could provide feedback on the utility and usability of the device
6. The extensive feedback obtained at each stage and the identification of key aspects of the activity, human user, and the contexts in which the device was used resulted in a product that ultimately met its intended outcome.

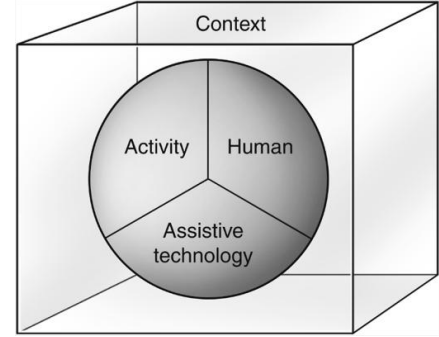
All components of the HAAT model have been considered.

Product research and development



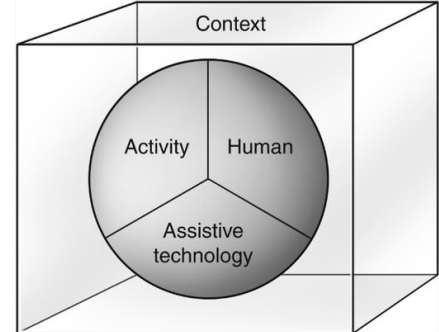
1. 收集家长对如厕产品需求，例如：不用的时候可以快速挪开，家里其他人也可以正常使用马桶。
2. 用户需求转化为设计特征，产出初代原型，同时产出原型评价量表。
3. 家长使用评价标准，对原型和市面上现有产品对比评估。
4. 同时让儿童试用原型，理疗师评估坐姿，儿童表达主观感受。
5. 产品定型前，家庭长时间实地使用，收集实用性、易用性反馈。
6. 每一轮充分收集反馈，综合人、活动、环境全部要素，最终产品达成预期目标。

Usability study



- While **utility** describes how well the device meets its intended function, **usability** describes how well the user is able to access the device's functionality
- Key features of usability:
 1. **Learnability**: All functions of the device are easily learned.
 2. **Efficiency**: The user meets the intended goals of device use in a reasonable amount of time with minimal frustration and effort.
 3. **Memorability**: How the device is used can be easily remembered, particularly when a function has not been used over a long period of time.
 4. **Errors**: This aspect refers to incorrect actions the user makes or actions that are omitted that limit or prevent a device from functioning as intended. It is important that errors can be recognized, that the effect of an error is minimal, that feedback is given to signal an error is made, and that the user can repair an error made.
 5. **Satisfaction**: The user has a positive experience when using the device, which we interpret as satisfaction with how the device functions as well as the image that the device use conveys to the user.
 6. **Ease of use**: This last criterion involves many of the previously mentioned ideas, but it also makes explicit the idea that a device must be simple to use on a regular basis, which we interpret as minimizing the number of steps required to generate the desired output.
- Two major perspectives of usability analysis are described:
 - In the first instance, the goal is to identify and rectify problems that a person encounters when using the device.
 - The second involves performing a number of tasks with the device and measuring the user's performance.
- Although both will provide useful information on problems with device use, the second type of analysis provides details on steps necessary to use the device as well as cognitive, communicative, sensory, and physical demands of device use.

可用性研究. Usability study



- 效用Utility：设备多大程度完成预设功能；
- 可用性Usability：使用者多大程度可以顺利使用设备的全部功能。

可用性六大关键指标：

1. 易学性：全部功能容易上手学习。
2. 效率：使用者可以在合理时间、较低精力消耗下完成目标，少产生挫败感。
3. 易记性：隔很久不用，也容易回忆操作方式。
4. 容错能力：识别误操作；错误带来的负面影响尽量小；提供错误反馈；使用者可以自行修正错误。
5. 主观满意度：使用者获得良好体验，对设备功能、自身使用设备的自我形象感到满意。
6. 易用性：日常操作简单，尽量减少操作步骤。

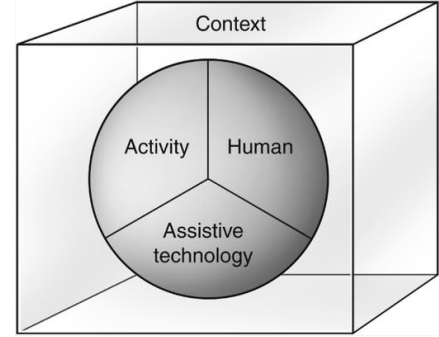
可用性分析两大思路：

- ① 找出使用者遇到的问题，加以修正；
- ② 完成一系列标准任务，测量使用者实际表现。

两种思路都有价值；第二种可以清晰看到操作步骤，以及设备对人认知、沟通、感官、体力提出的负担。

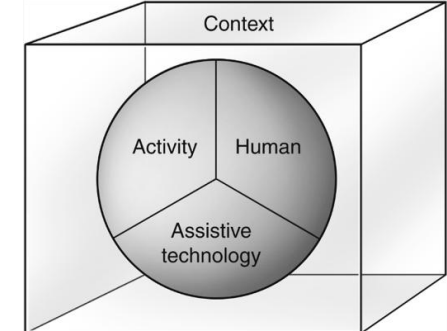
Clinical assessment

临床评估



- Basic process:
 - identification of the need to be addressed by AT use,
 - assessment of key aspects,
 - synthesis of the assessment results,
 - device recommendation.
- Assessment is user centered:
 - the user of AT—and others as appropriate such as a parent, child, or spouse—are equal partners on the assessment team and that their goals for device use are paramount in driving the assessment process.
- Assessment can include both formal and informal instruments:
 - formal instruments: tests developed using rigorous procedures with established measurement properties, administered and interpreted in a standardized manner
 - informal assessment: observation and interview to gather input and information.
- The clinical team includes the AT users (and others, such as family members), audiologists, occupational and physical therapists, speech language pathologists, nurses, physicians, teachers, rehabilitation **engineers**, and rehabilitation and educational assistants.
- Institutional policies and practices determine who is present during an assessment and what their responsibilities are regarding different aspects of the assessment.
- Assessment culminates in device recommendation and acquisition.
- When possible, the assessment process also results in procurement of funding. This component requires knowledge of who is responsible to request funding and what is required to make that request.

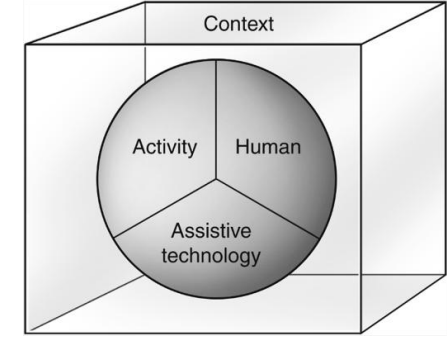
Clinical assessment



基础流程：识别辅助技术需求 → 多维度评估 → 整合评估结果 → 给出设备推荐。

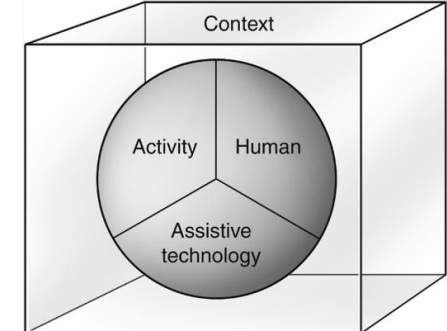
- 以用户为中心：AT设备使用者本人（必要时家属父母配偶）是评估团队平等参与者；使用者目标优先级最高。
- 评估工具：标准化正式测评工具；非正式手段（观察、访谈）。
- 评估团队：AT设备使用者与家属、听力学医师、作业治疗师、物理治疗师、言语治疗师、护士、医生、教师、康复工程师、康复助理。
- 制度政策决定参与人员与分工；评估最终输出设备选型，尽可能同步申请经费支持。

Outcome evaluation



- Two aspects:
 - Evaluation of the outcome of device use by an individual client, at several points after acquisition of the device: immediately after acquisition and short- and long-term follow-up.
 - Outcome of device use for a group of individuals.
- Outcome evaluation is based on the goals identified by the client regarding the acquisition of the device. This evaluation determines how well these goals have been met.
- In addition, other outcomes, such as satisfaction with the device and the psychosocial impact of the device use can also be incorporated into the individual client evaluation
- The HAAT model was used to interpret data in a study of children's perceptions of their use of writing and communication aids. The HAAT model was useful to identify barriers and enablers to the use.
- On a larger scale, evaluation of the outcomes of device use with a sample of clients provides information on device use, the ability of the device to meet the intended need, and feedback on device design for future development. Commonly, this type of evaluation is conducted as part of a formal research study with the purpose of building an evidence base related to the AT use. This evidence base supports clinical decisions, influences policy, and justifies new and continued public funding of assistive devices and services.
- The HAAT model and associated process will benefit from continued development of more specific processes and methods to support outcome evaluation research

使用效果评价 Outcome evaluation



4) 使用效果评价

两层含义：

- ① 个体使用者：拿到设备后即时、短期、长期多次随访，评估使用效果。依据使用者当初设定的目标，判断目标达成情况；同时包含满意度、设备带来的心理社会影响。
- ② 群体层面：一批使用者整体使用结果。

HAAT模型被用于儿童书写沟通辅助工具研究，识别使用促进因素与阻碍。群体评估为产品设计提供反馈，产出循证证据，支撑临床决策、政策制定，论证公共经费投入合理性。

HAAT相关流程仍有待开发更细化的方法工具，服务效果评估研究。

Universal design

BOX 2.1 Principles of Universal Design

1. Equitable use: The design is useful and marketable to people with diverse abilities.
2. Flexibility in use: The design accommodates a wide range of individual preferences and abilities.
3. Simple and intuitive use: Use of the design is easy to understand regardless of the user's experience, knowledge, language skills, or current concentration level.
4. Perceptible information: The design communicates necessary information effectively to the user regardless of ambient conditions or the user's sensory abilities.
5. Tolerance for error: The design minimizes hazards and the adverse consequences of accidental or unintended actions.
6. Low physical effort: The design can be used efficiently and comfortably and with a minimum of fatigue.
7. Size and space for approach and use: Appropriate size and space is provided for approach, reach, manipulation, and use regardless of user's body size, posture, or mobility.

Erroneously intended as “a product for all”

It is a philosophy that encourages designers to consider the needs of the largest possible range of users

It typically brings to products for the largest possible citizens range, not necessarily for all

- “Universal Design” (in USA and Japan)
- “Design for All” (in Europe)
- “Inclusive Design” (in UK)

- “... designers ensure that their products and services address the needs of the widest possible audience”

Definition by the British Ministry of Trade and Industry, 2000

- “... Universal design is an approach to design that incorporates products as well as building features which, to the greatest extent feasible, can be used by everyone”

Definition by universal design pioneer Ron Mace, 1985

通用设计七大原则.

Universal design principles:

1. Equitable use ① 公平使用

Principle 1: Equitable Use

The design is useful and marketable to people with diverse abilities.

Guidelines:

- 1a. Provide the same means of use for all users: identical whenever possible; equivalent when not.
- 1b. Avoid segregating or stigmatizing any users.
- 1c. Provisions for privacy, security, and safety should be equally available to all users.
- 1d. Make the design appealing to all users.

1. 公平使用：设计对能力各不相同的人群都具备使用价值，具备市场价值。

指南：

尽可能提供完全一致的使用方式，条件不允许则提供等效方案；
不隔离、不标签化特定用户；
所有人同等保障隐私、安全；
设计对全部使用者具备吸引力。



Universal design principles:

2. Flexibility in use ② 灵活使用

Principle 2: Flexibility in Use

The design accommodates a wide range of individual preferences and abilities.

Guidelines:

- 2a. Provide choice in methods of use.
- 2b. Accommodate right- or left-handed access and use.
- 2c. Facilitate the user's accuracy and precision.
- 2d. Provide adaptability to the user's pace.

原则2：使用弹性（Flexibility in use）

设计应适配广泛的个人偏好与能力。

指导细则：

- 2a. 提供多种使用方式供选择。
- 2b. 适配左手或右手操作使用。
- 2c. 助力用户提升操作的准确度与精度。
- 2d. 支持适配用户自身的操作节奏。



Universal design principles:

3. Simple and intuitive use

③ 简单直观.

无论用户的经验、知识、语言能力或是当下的注意力水平如何，该设计都易于理解。

Principle 3: Simple and Intuitive Use

Use of the design is easy to understand, regardless of the user's experience, knowledge, language skills, or current concentration level.

Guidelines:

- 3a. Eliminate unnecessary complexity.
- 3b. Be consistent with user expectations and intuition.
- 3c. Accommodate a wide range of literacy and language skills.
- 3d. Arrange information consistent with its importance.
- 3e. Provide effective prompting and feedback during and after task completion.

指导细则：

- 3a. 消除不必要的复杂结构。
- 3b. 与用户的预期和直觉保持一致。
- 3c. 适配广泛的读写能力与语言水平。
- 3d. 按照信息的重要性排布信息。
- 3e. 在任务进行过程中和任务完成后，提供有效的提示与反馈。



<https://universaldesign.ie/What-is-Universal-Design/The-7-Principles/>

④ 信息可感知.

Universal design principles:

4. Perceptible information

无论环境条件、或是用户感官能力存在差异，该设计都能向用户有效传递必要信息。

Principle 4: Perceptible Information

The design communicates necessary information effectively to the user, regardless of ambient conditions or the user's sensory abilities.

Guidelines:

- 4a. Use different modes (pictorial, verbal, tactile) for redundant presentation of essential information.
- 4b. Provide adequate contrast between essential information and its surroundings.
- 4c. Maximize "legibility" of essential information.
- 4d. Differentiate elements in ways that can be described (i.e., make it easy to give instructions or directions).
- 4e. Provide compatibility with a variety of techniques or devices used by people with sensory limitations.

Lighting accentuates train doorways and the adjoining gates prevent riders from falling onto the tracks



- 4a. 使用多种方式（图像、文字、触觉）对关键信息进行冗余呈现。
- 4b. 在核心信息与其背景环境之间提供充足的对比度。
- 4c. 最大化关键信息的易读性。
- 4d. 用便于描述的方式区分不同元素（例如：便于给出指令或指引）。
- 4e. 兼容感官障碍人群所使用的各类辅助技术或设备。

⑤ 容错性.

Universal design principles:

5. Tolerance for error

设计应最大限度降低意外操作或无意操作带来的危险与不良后果。

Principle 5: Tolerance for Error

The design minimizes hazards and the adverse consequences of accidental or unintended actions.

Guidelines:

- 5a. Arrange elements to minimize hazards and errors: most used elements, most accessible; hazardous elements eliminated, isolated, or shielded.
- 5b. Provide warnings of hazards and errors.
- 5c. Provide fail safe features.
- 5d. Discourage unconscious action in tasks that require vigilance.

- 5a. 布局元素以降低危险与错误：将最常用的元素放在最易取用的位置；消除、隔离或防护危险部件。
- 5b. 针对危险与错误提供警示。
- 5c. 设置故障安全保护机制。
- 5d. 在需要保持警惕的任务中，避免无意识误操作。



⑥ 降低体力消耗.

Universal design principles: 6. Low physical effort

Principle 6: Low Physical Effort

The design can be used efficiently and comfortably and with a minimum of fatigue.

Guidelines:

- 6a. Allow user to maintain a neutral body position.
- 6b. Use reasonable operating forces.
- 6c. Minimize repetitive actions.
- 6d. Minimize sustained physical effort.

- 6a. 允许用户保持身体自然放松的中立姿态。
- 6b. 使用合理的操作力度。
- 6c. 尽量减少重复动作。
- 6d. 降低持续发力的体力消耗。



⑦ 适配尺寸与操作空间.

Universal design principles:

7. Size and space for approach and use

无论使用者的体型、姿态或移动能力如何，都提供合适的尺寸与空间，供使用者靠近、伸手触及、操作与使用。

Principle 7: Size and Space for Approach and Use

Appropriate size and space is provided for approach, reach, manipulation, and use regardless of user's body size, posture, or mobility.

Guidelines:

- 7a. Provide a clear line of sight to important elements for any seated or standing user.
- 7b. Make reach to all components comfortable for any seated or standing user.
- 7c. Accommodate variations in hand and grip size.
- 7d. Provide adequate space for the use of assistive devices or personal assistance.

- 7a. 无论使用者是坐姿还是站姿，都能清晰看到关键部件。
- 7b. 让坐姿或站姿的使用者，都能舒适地触及全部组件。
- 7c. 适配手掌大小与抓握尺寸的差异。
- 7d. 预留足够空间，方便使用辅助器具或获得他人协助。



AT 的可接受性

Acceptability of assistive technology

理性因素 (Rational factors) : 感知有用性、有效性、效率、可靠性、稳健性

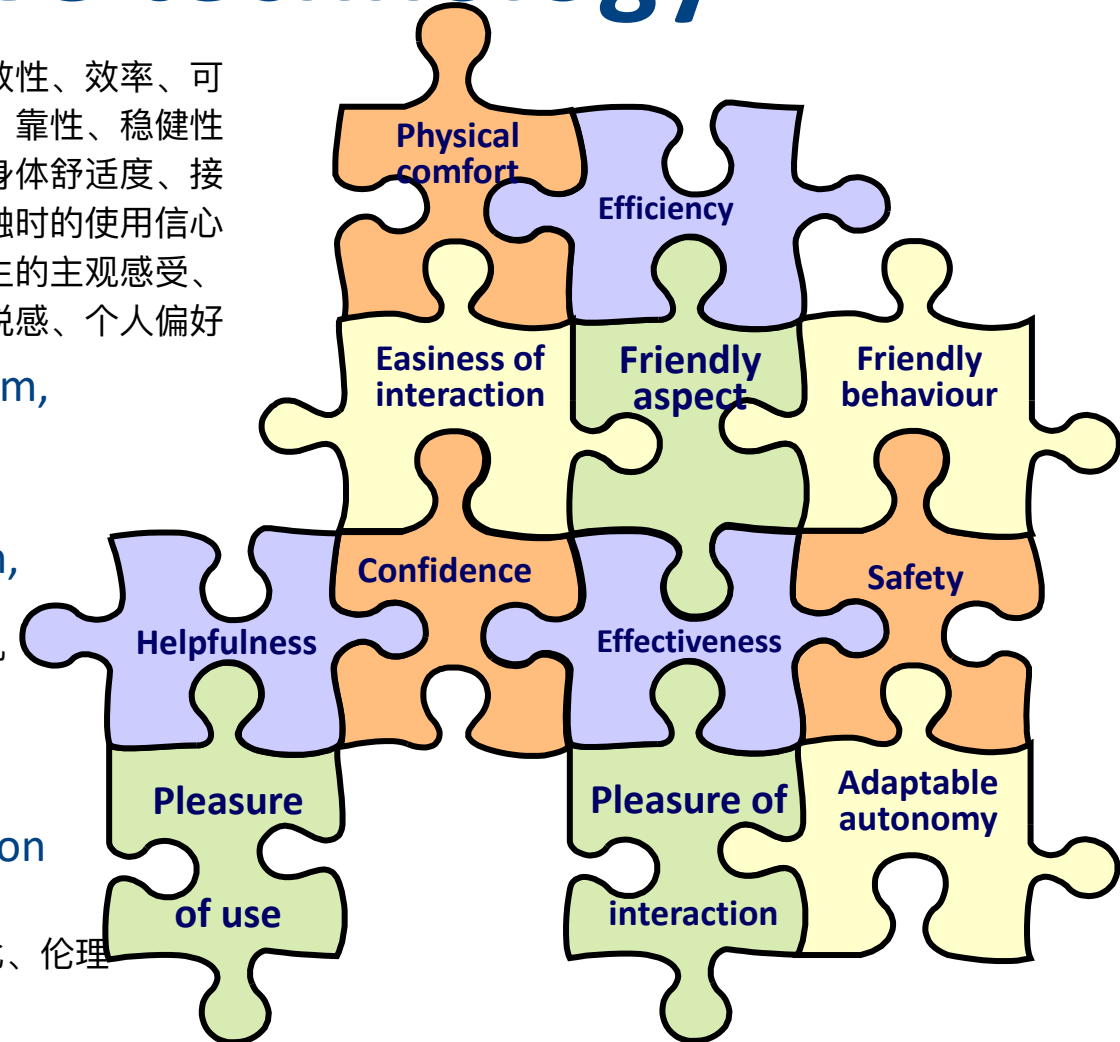
- **Rational factors:** 人体工学因素 (Ergonomic factors) : 交互时的身体舒适度、接
perceived helpfulness, effectiveness, efficiency, reliability, 触时的使用信心
robustness

- **Ergonomic factors:** 情绪因素 (Emotional factors) : 与系统交互产生的主观感受、
使用过程中的愉悦感、个人偏好
perceived physical comfort, in the interaction with the system,
confidence in contact

- **Emotional factors:** positive emotional feeling in the interaction with the system,
pleasure of use and interaction, liking aspects

- **Psychological factors:** 心理因素 (Psychological factors) : 用户与机
器人之间积极的关系、合适的机器人自主程度、
尊重用户偏好与隐私、交互的易用性
positive relationship between the user and the robot
personalities, proper balance of robot autonomy, respect of
user's preferences and privacy, easiness of use and interaction

- **Economic and social factors:** 经济与社会因素
Cost/benefit ratio, Cost performance ratio, 成本收益比、性价比、伦理
Ethical aspects, Healthcare Policy 层面、医疗政策



用户中心设计思路.

User-Centred Design (UCD)

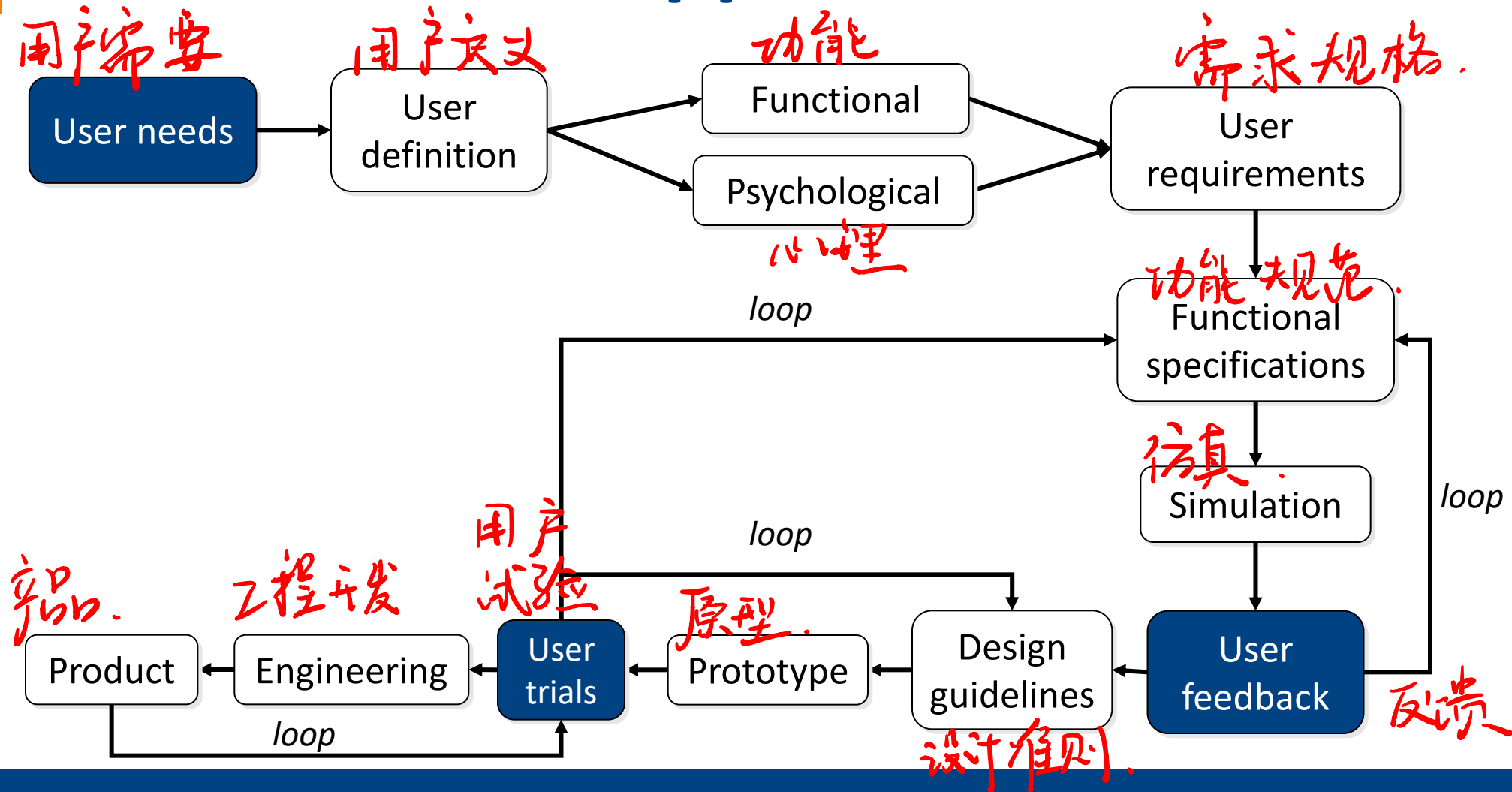


逻辑链：UCD → 用户可接受度 → 辅助技术（AT）的广泛落地应用

Involving users in the design process
Users = individuals with disability,
clinical staff, caregivers, family members

在设计流程中吸纳用户参与
用户 = 残障人士、临床工作人员、照护者、家属

面向用户的设计思路. User-oriented approach



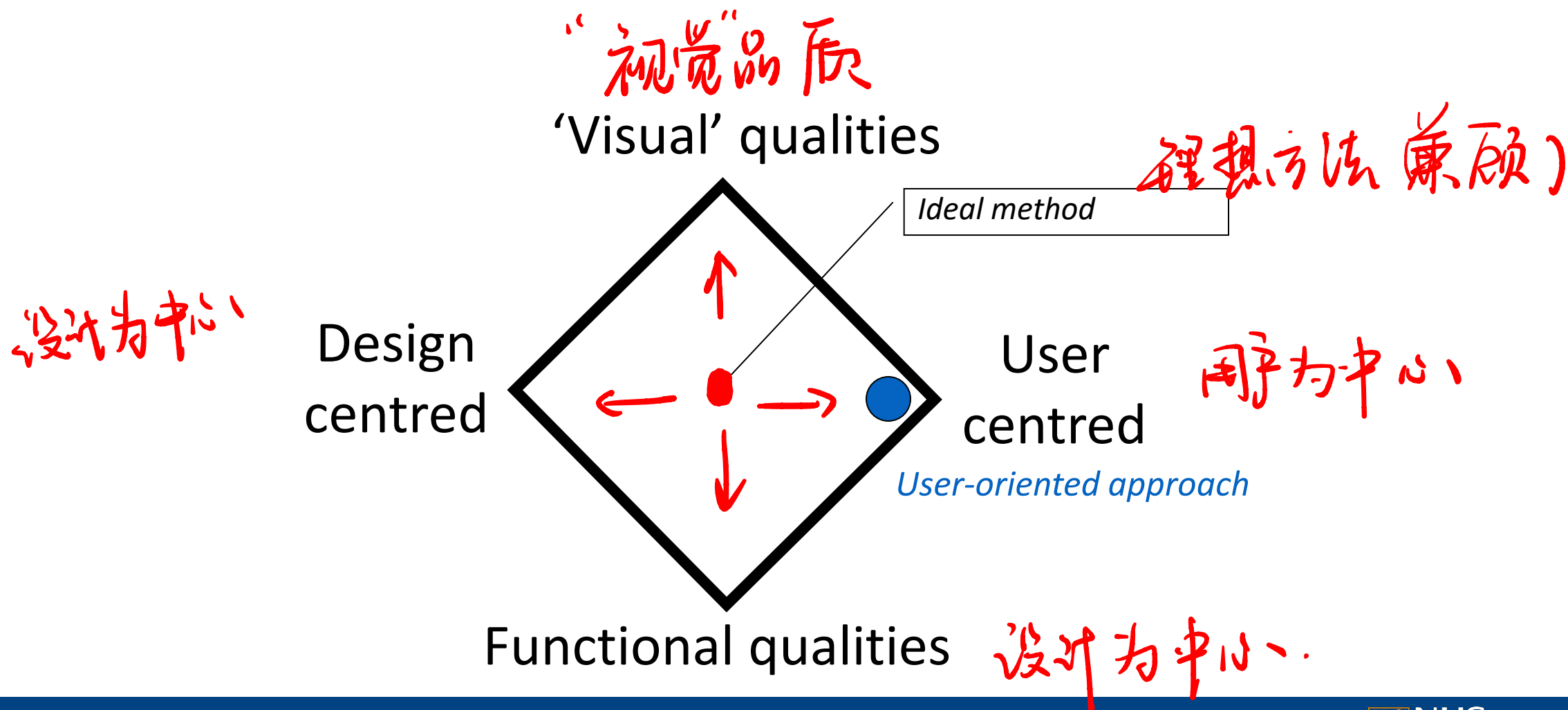
Possible limitations of the user-oriented approach

潜在局限。

- Complex, time-demanding
- End users can hardly imagine radically innovative products
- They can only express incremental innovation
- Thus, the user-oriented approach **may limit innovation**
 - 流程复杂，耗时较长
 - 终端用户很难设想颠覆性创新产品
 - 用户仅能提出渐进式改良创新
 - 因此，面向用户的设计方法可能会限制创新

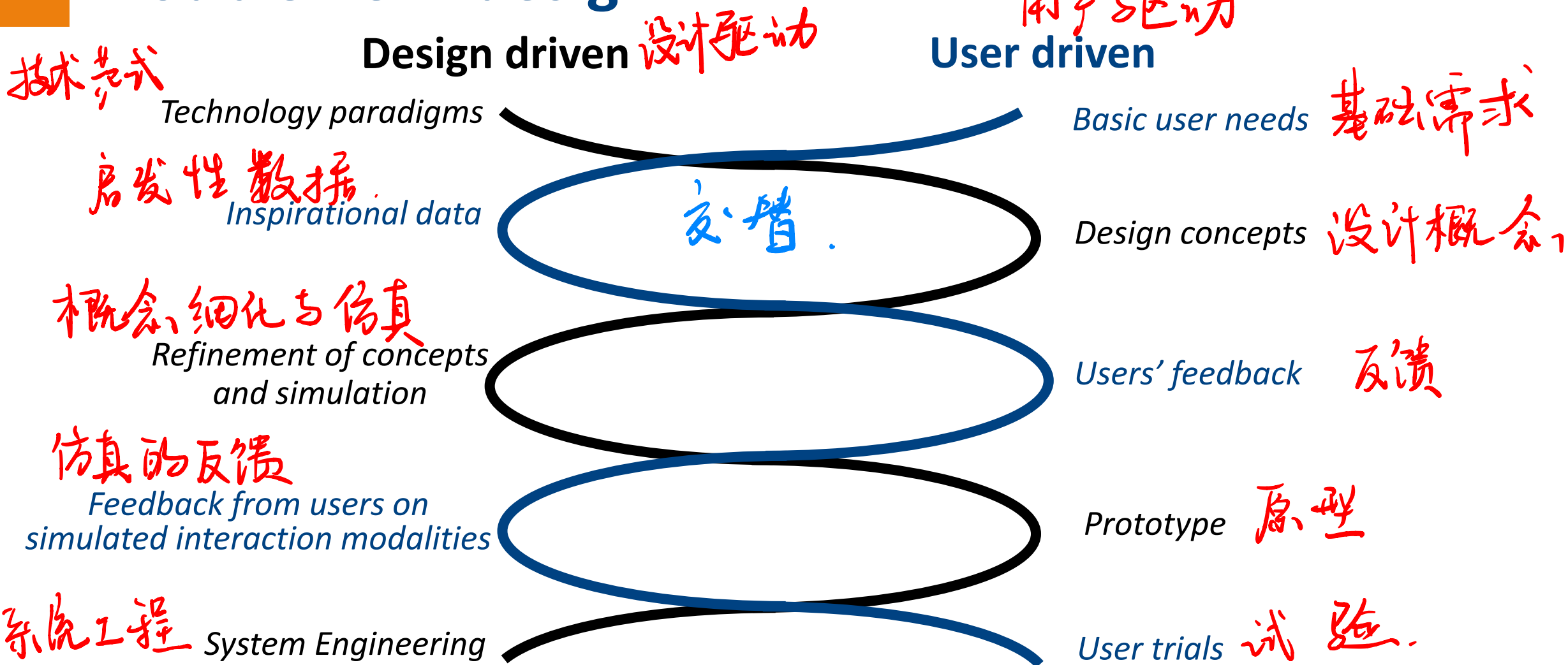
设计方法四方图.

The square of design methods



双螺旋设计.

Double helix design



辅助技术的特征分类

Characterization of Assistive Technologies (AT)

- Assistive vs. rehabilitative or educational technologies
- Low to high tech technology
- Hard technologies and soft technologies
- Appliances vs. tools
- Minimal to maximal technology
- General vs. specific technologies
- Commercial to custom technology

分类维度

- 辅助技术 vs 康复技术 / 教育技术
- 从低技术到高技术
- 硬技术与软技术
- 设备 vs 工具
- 最低限度技术到最大化技术
- 通用技术 vs 专用技术
- 商用技术 vs 定制化技术

Assistive vs. rehabilitative or educational technologies 辅助技术 vs 康复/教育技术

辅助技术 (Assistive technologies)：满足各类功能需求，帮助使用者完成功能性活动。

- **Assistive technologies** serve a variety of functional needs. They help an individual to carry out a functional activity
- **Rehabilitative and educational technologies** are used within a rehabilitative or educational process, respectively, and are used as tools for remediation or rehabilitation, rather than being a part of the person's daily life and functional activities. Often they are used for developing skills for the use of assistive technologies

康复与教育技术：分别应用于康复或教育流程中，作为矫正、康复的工具，并不作为用户日常生活与功能性活动的组成部分；常用来培养使用者操作辅助技术所需的技能。

从低技术到高技术 (Low to high tech technology)

Low to high tech technology

- 低技术 (Low-tech) : 制作简单、容易获取

例: 简易纸笔沟通板、改良餐具、简易夹板

Not fully precise distinction

- **Low-tech:** simple to make and easy to obtain

- e.g., simple pencil-and-paper communication boards, modified eating tools, simple splints

- 高技术 (High-tech) : 价格昂贵, 制作与获取难度更高

例: 电动轮椅、电子通讯设备、计算机辅助助听设备、辅助机器人

- **High-tech:** expensive, more difficult to make and harder to obtain

- e.g., wheelchairs, electronic communication devices, computer-based aids, assistive robots



硬技术与软技术 (Hard technologies and soft technologies)

Hard technologies and soft technologies

硬技术 (Hard)：有形、可直接获取的零部件，可组装成辅助技术系统。

- **Hard:** tangible, readily available components that can be assembled into assistive technology systems
- **Soft:** human area of decision making, strategies, training, concept formation, and service delivery. They are generally captured as:
 - 软技术 (Soft)：人的决策环节、方法策略、培训、概念构建、服务交付。一般可归纳为：
 - 人员
 - 书面资料
 - 计算机程序
 - People
 - Written
 - Computer

These aspects of technology are much harder to obtain, but without them the hard technologies cannot be successful.

They are difficult to acquire because they are highly dependent on human knowledge rather than on tangible objects

软技术更难获取；缺少软技术，硬技术无法发挥作用。

软技术难以掌握，因为它高度依赖人类知识，而非实体物件。

Practice, Mosby Elsevier, 3rd Edition, 2008

设备 vs 工具 (Appliances vs. tools)

Appliances vs. tools

- **Appliances** provide benefit to an individual independently of the individual's skill level
 - e.g., refrigerator
- **Tools** require the development of skill for their use
 - e.g., hammer
 - 设备 (Appliances) : 无论使用者技能水平高低, 都能独立为使用者带来益处。
例: 冰箱
 - 工具 (Tools) : 使用者需要掌握相应技能之后, 才能使用它。
例: 锤子

最低限度技术到最大化技术

Minimal to maximal technology

基于用户需求的连续谱系：

Continuum of user's needs



Devices that provide some assistance or augment the individual's ability to perform a task

提供部分辅助、增强用户执行任务能力的设备

Assistive technologies that replace a significant amount of ability to generate functional outcome

替代大量能力、产生功能性结果的辅助技术

- Minimal technologies improve functionality
 - e.g., orthoses, limited-use wheelchairs (e.g., in airports or public spaces)
- Maximal technologies replace functions
 - e.g., prostheses, wheelchairs, communication aids, ...

最低限度技术：改善原有功能

例：矫形器、限时使用轮椅（如机场/公共场所临时轮椅）

最大化技术（Maximal technologies）：直接替代原有功能

例：假肢、轮椅、沟通辅助器具

通用技术 vs 专用技术

General vs. specific technologies

通用技术 (General-purpose technologies) : 可应用于多种不同场景

例子: 坐姿与定位系统、控制交互界面、计算机

- **General-purpose technologies** are used in many different applications

- e.g., seating and positioning systems, control interfaces, computers

- **Specific technologies** are intended for a specific application

专用技术 (Specific technologies) : 专为某一类特定用途设计

例子: 通讯设备、轮椅、喂食设备、助听器、视障人群移动辅助器具

- e.g., communication devices, wheelchairs, feeding devices, hearing aids, mobility aids for people with visual impairments



MicroSoft
stock images

商品化技术到定制化技术

Commercial to custom technology



Mainstream
technology

主流通用



Commercially available
assistive technology

市场辅助力



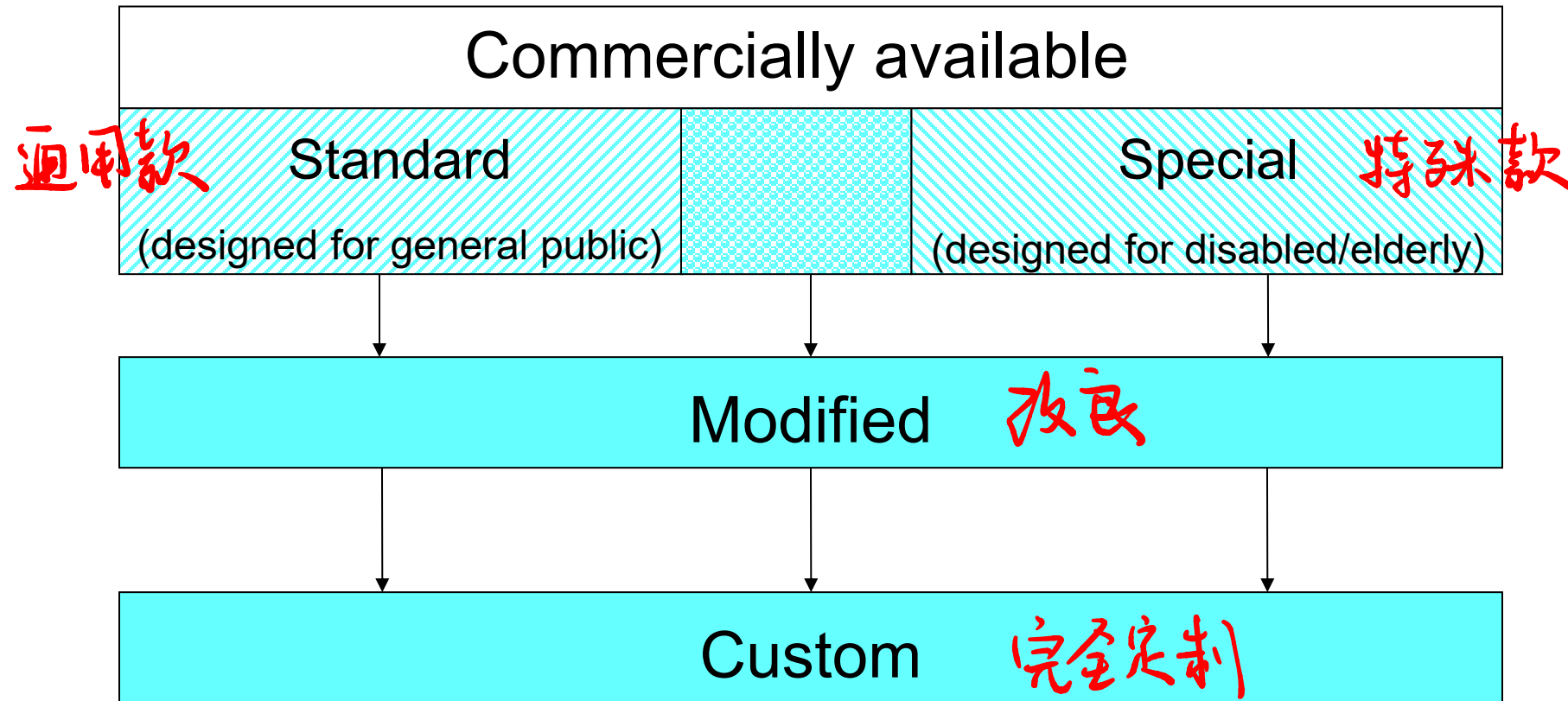
Custom-made
assistive technology

定制化
辅助

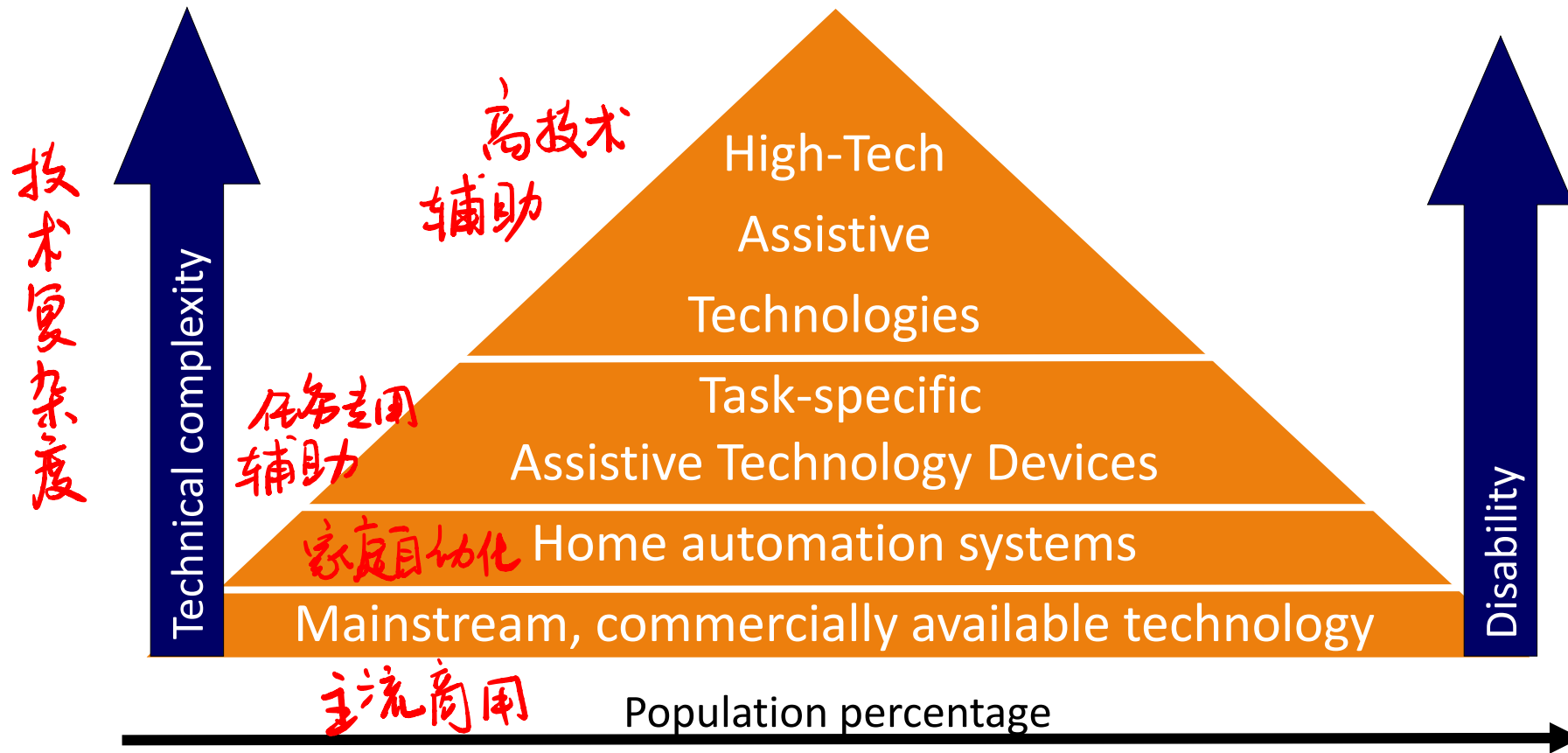
Mass-produced devices are typically easier to obtain and less expensive than devices that are produced specifically for individuals with disabilities

量产设备通常更容易获取，并且比专门为残障人士单独制作的设备成本更低。

Commercial to custom technology



Assistive technology pyramid



机器人技术 AT

Robotics as Assistive Technology (AT)

机器人分类

Robot Classification (ISO 2012)

- 工业机器人 (Industrial robot) : 自动控制、可重新编程、多用途操作器, 具备3个及以上运动轴; 可固定安装或移动, 用于工业自动化场景。
- 服务机器人 (Service robot) : 为人类或设备执行有用任务的机器人, 不含工业应用场景。

服务机器人分为两类:

- **Industrial robot:** “automatically controlled, reprogrammable, multipurpose manipulator, programmable in three or more axes, which can be either fixed in place or mobile for use in industrial automation applications”
- **Service robot:** “robot that performs useful tasks for humans or equipment excluding industrial applications”
- Service robots can be for
 - 1. 专业型 (Professional use) : 用于商业任务, 通常由受过专业培训的操作人员使用。
 - **Professional use, used for a commercial task, usually operated by a properly trained operator**
 - 2. 个人型 (Personal use) : 用于非商业任务, 一般由普通使用者操作。
 - **Personal use, used for a non-commercial task, usually by laypersons**

Personal Service Robots

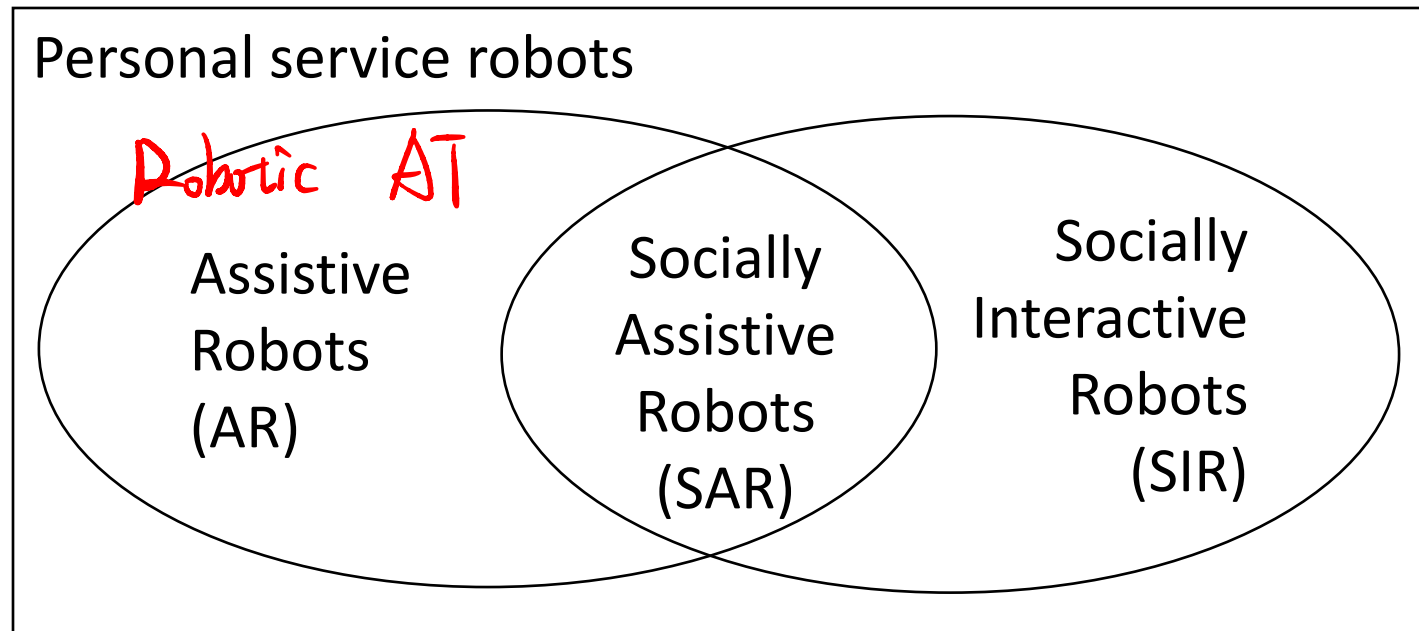
个人服务机器人 (Personal Service Robots)

韦恩图三个分支：

1. 辅助机器人 (Assistive Robots, AR)
2. 社会辅助机器人 (Socially Assistive Robots, SAR)
3. 社会交互机器人 (Socially Interactive Robots, SIR)

定义：

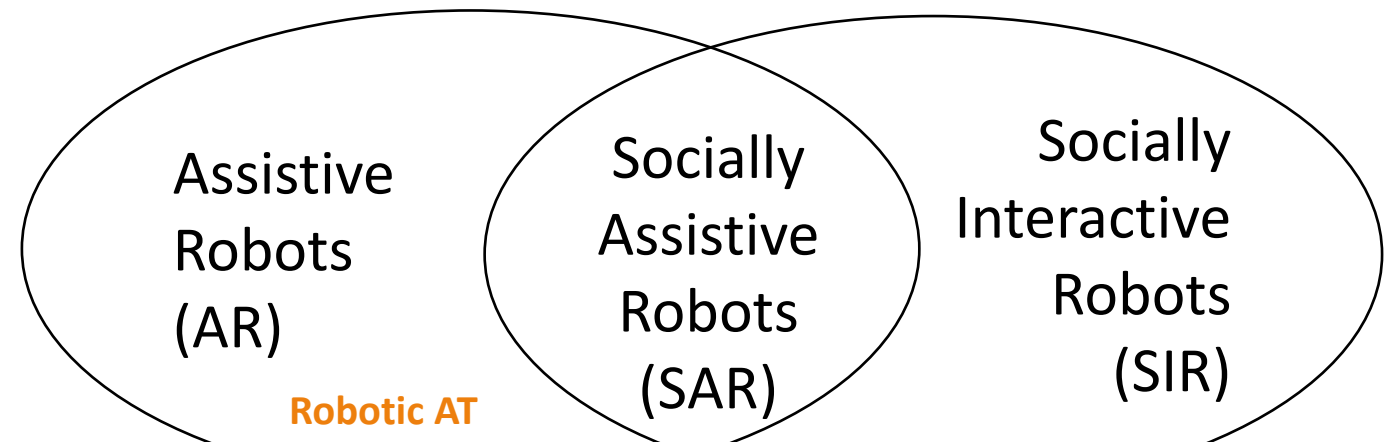
- 辅助机器人(AR)： 为人类使用者提供辅助或支持
- 社会交互机器人(SIR)： 社会交互是其核心功能
- 社会辅助机器人(SAR)： 通过社交交互辅助用户
- 物理辅助机器人： 在肢体层面协助用户



- **Assistive Robots (AR)** give aid or support to a human user
- In **Socially Interactive Robots (SIR)** social interaction plays a key role
- **Socially Assistive Robots (SAR)** assist the user with social interaction
- **Physically Assistive Robots** assist the user with physical interaction

Robotic AT (Assistive Technologies)

Personal service robots



Robotic AT (机器人辅助技术) 属于辅助机器人AR的一个子类
机器人辅助技术供残障人士使用 (而普通辅助机器人AR可面向任意使用者)

A sub-class of Assistive Robots (AR), because they are used by individuals with disabilities (while AR are used by any user)

- Robotic prosthetic arms and legs
 - Robotic manipulators
 - Bionic arms
 - Exoskeletons
- 机器人辅助技术实例:
- 机器人假肢手臂与腿
 - 机器人操作器 / 机械臂 (Robotic manipulators)
 - 仿生手臂 (Bionic arms)
 - 外骨骼 (Exoskeletons)



机器人辅助技术的演进.

Evolution of Robotic Assistive Technologies

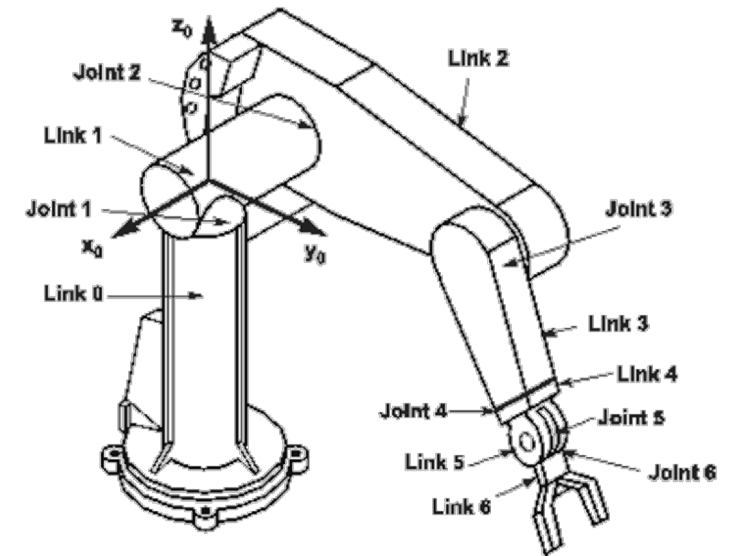
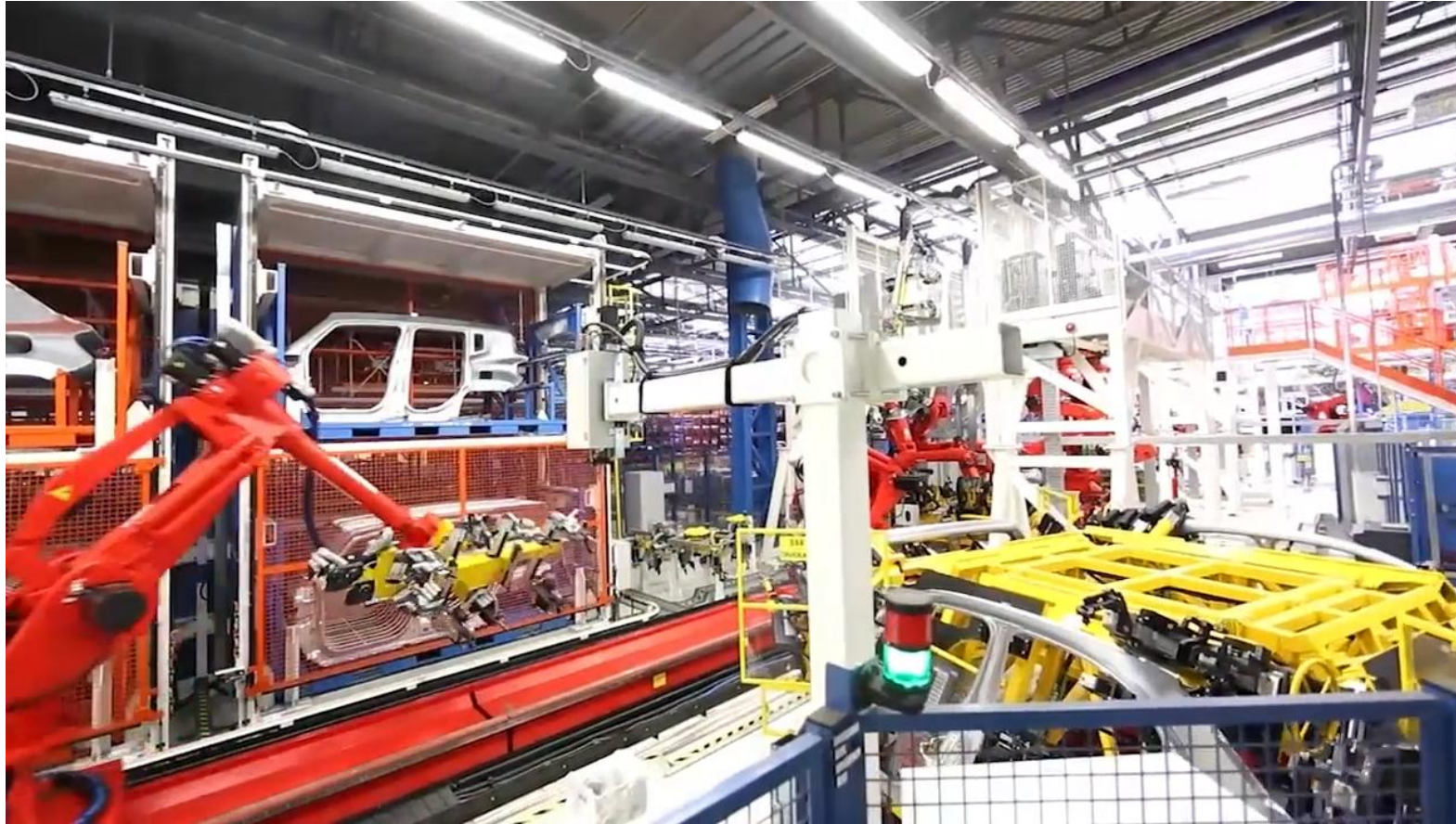
1. 工业机器人 (Industrial Robots)

- 环境特点：结构化环境 (Structured environments)
- 部署形式：机器人固定在环境中

Industrial Robots

多用于工厂自动化场景，工作环境预先设定，场景固定。

Structured environments
Robot fixed in the environment



ROBOT

Reliability (minimal requested values):
Mean Time Before Failure = 40,000 hrs
Efficiency $\eta > 99.99875\%$ (Source: COMAU)

Fixed Assistive Robots

环境特点：结构化环境 (Structured environments)

• 部署形式：机器人固定在环境中

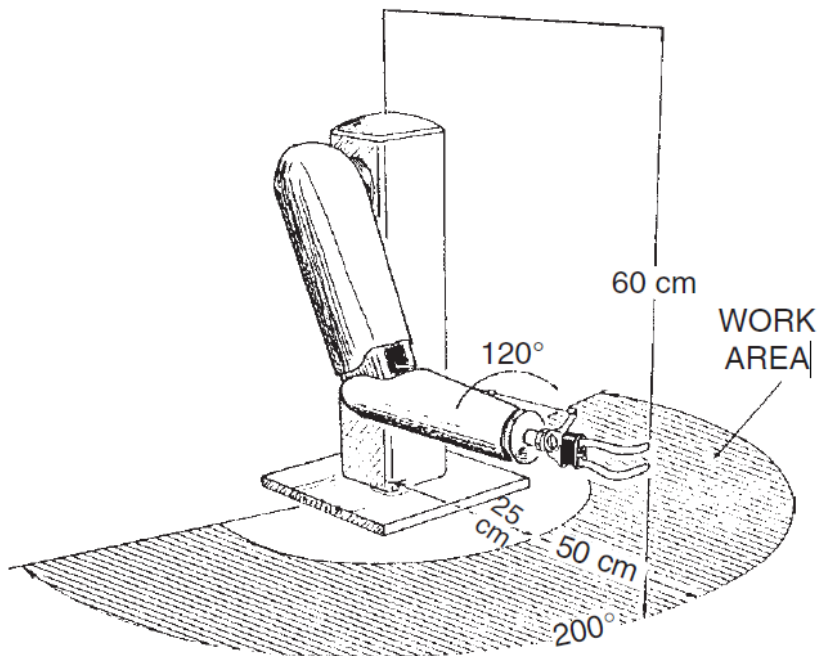
代表系统：

1. APL RAWT system
2. Master-Raid
3. DeVAR

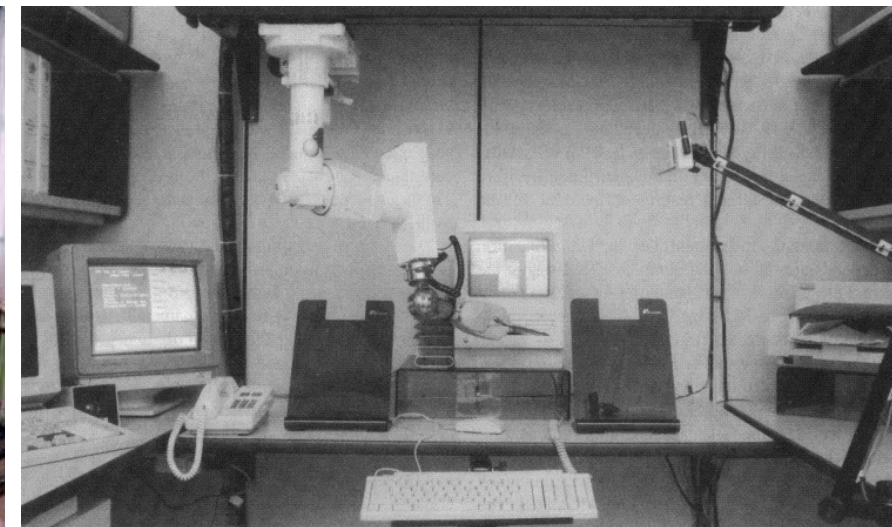
APL RAWT system

Master-Raid

DeVAR



备注：工厂以外使用的机器人需要应对更复杂、非结构化的环境。



Track-mounted upside-down PUMA-260 with Otto-Bock gripper, for manipulation of objects in a vocational desktop environment

服务机器人 Service Robots

Robots outside factories need to negotiate **unstructured environments** and react to unexpected situations



- **Navigation**

工厂之外的机器人，需要在非结构化环境（unstructured environments）中自主通行，并且能够应对突发、未知场景。
两大核心能力：

- Navigation 导航

- Mobile manipulation 移动操作

- **Mobile manipulation**



<https://youtu.be/BiemOdLYf48>

Smart Wheelchairs

1. 智能轮椅

搭载导航传感器与导航算法，实现自主移动。



Navigation sensors and algorithms



Wheelchair-mounted Manipulators

轮椅搭载式操作机械臂

机械臂安装在轮椅本体上，辅助用户完成抓取、操作物体。



移动式辅助机器人
例：Care-o-bot

Mobile Assistive Robots



Care-o-Bot, Fraunhofer Institute, Stuttgart, Germany



TIDE-MOVAID, 1997,
Scuola Superiore Sant'Anna,
Pisa, Italy



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<https://www.argallab.northwestern.edu/research/robot-platforms/robotic-wheelchair/>

<https://www.robopreneur.com/roboticwheelchair>

以用户为中心的设计在洗澡辅助机
器人上的应用案例

Example of Application of User-centred Design to an Assistive Robot for Showering

I-Support

I-Support:

I-Support: 用于洗浴任务的服务机器人系统

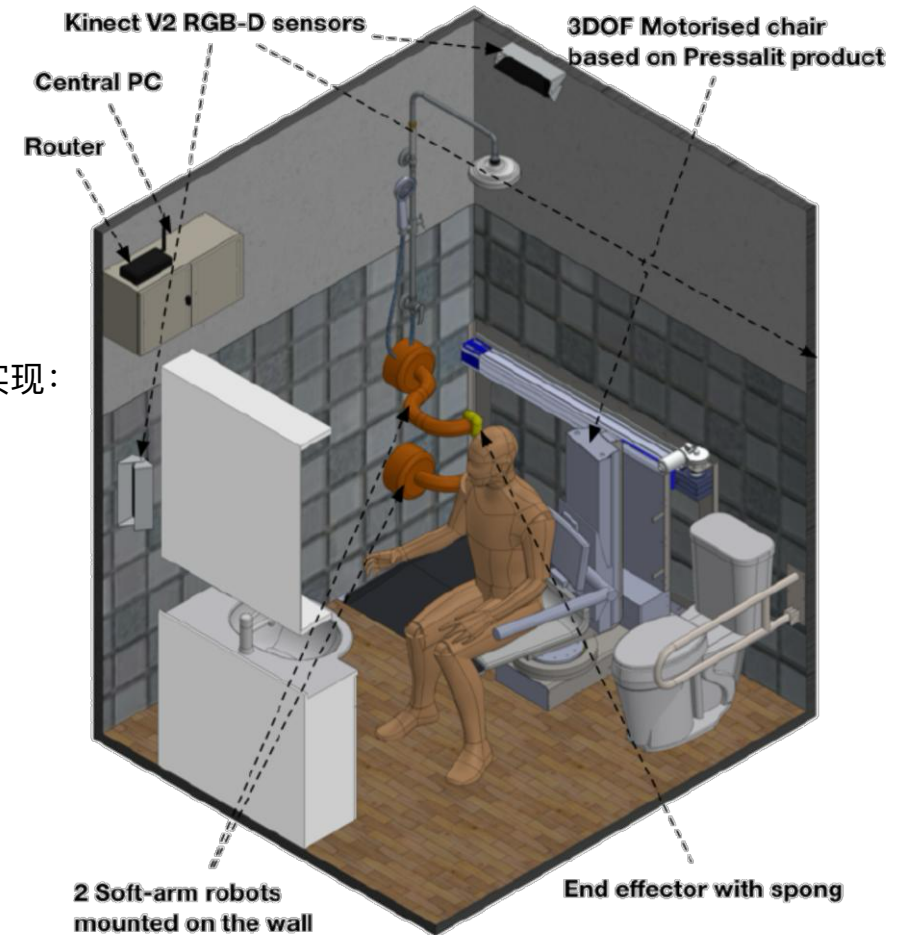
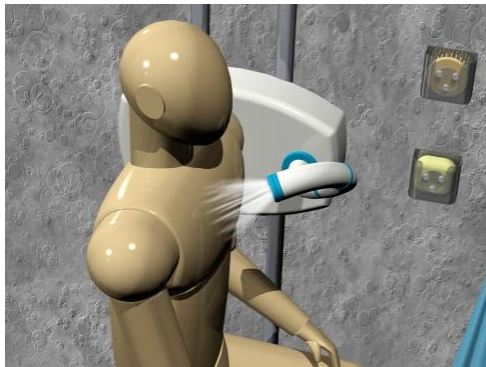
A Service Robotics System for Bathing Tasks

目标

- Objectives:
- 保障使用者自主性与隐私
 - 减轻护理人员的高阶工作负担
 - Autonomy and privacy for the user
 - Higher-level assistance for the carer

系统组成

- A motorized shower chair
- A soft robotic shower arm, for:
 - pouring water and detergent
 - wiping and scrubbing
- 电动洗浴椅
- 柔性机器人手臂，可实现:
 - 倾倒水与沐浴液
 - 擦拭与擦洗



用户为中心的设计 -

User-Centred Design (UCD)

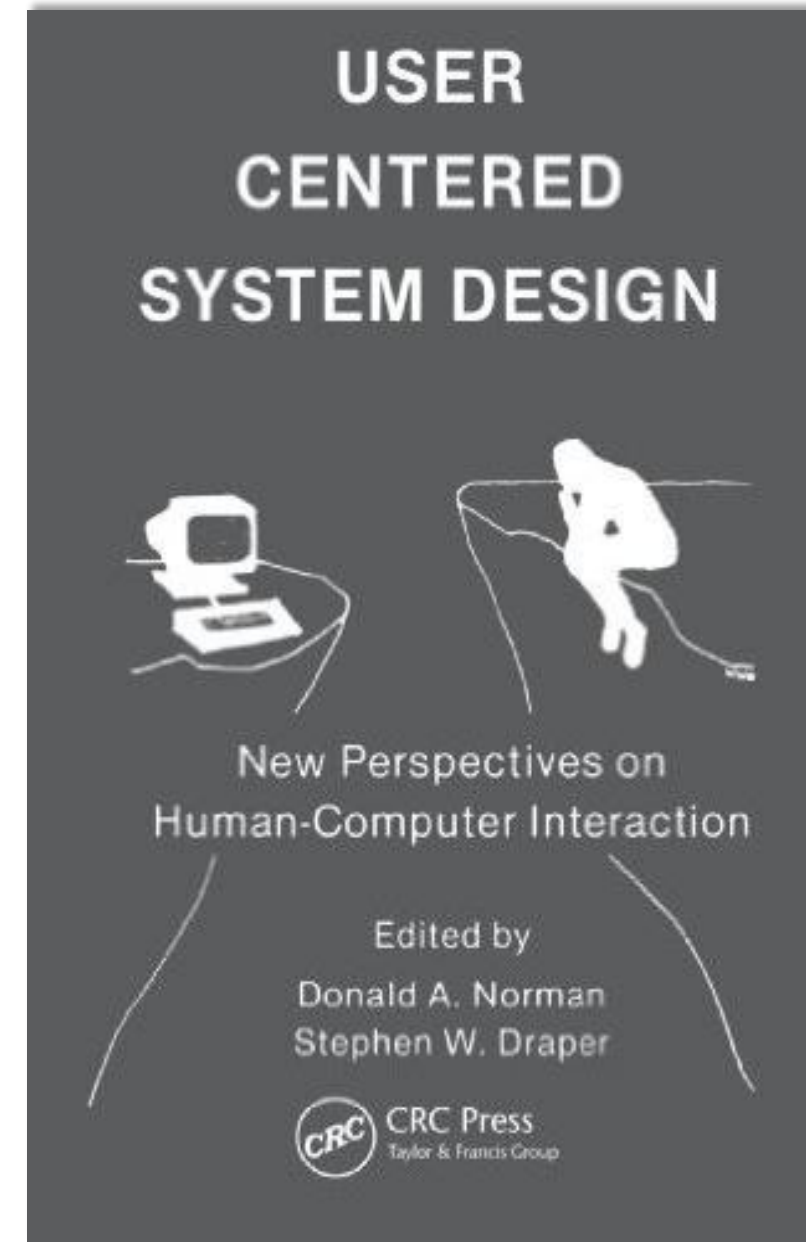
Definition of the **SCENARIO** and search for **END-USERS**



Definition of **USER NEEDS** through interviews, focus groups, surveys

UCD流程:

1. 定义场景 (SCENARIO) , 寻找终端用户
2. 通过访谈、焦点小组、问卷调研, 挖掘用户需求 (USER NEEDS)



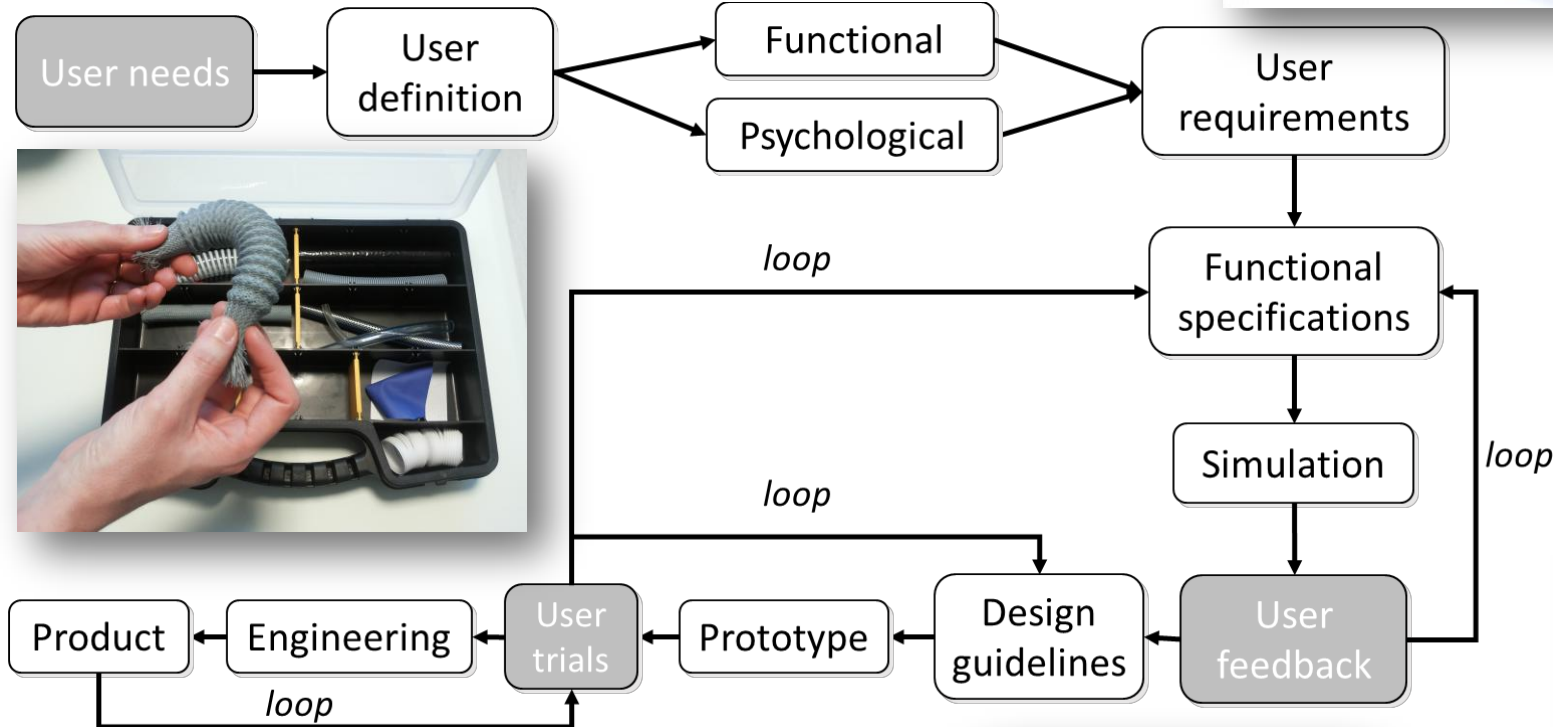
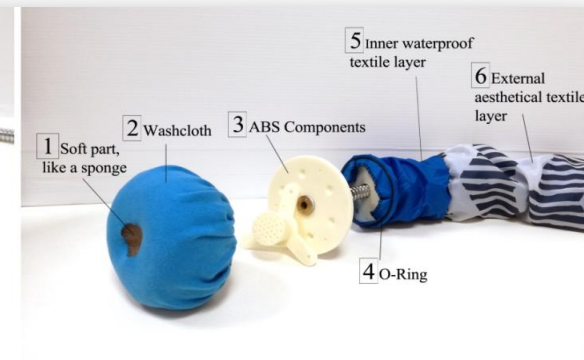
Norman, D. A., & Draper, S. W. (1986). *User centered system design*. Hillsdale, NJ.

流程 (前面讲过)

User-oriented Design

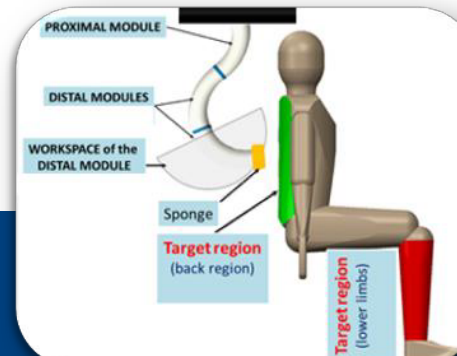
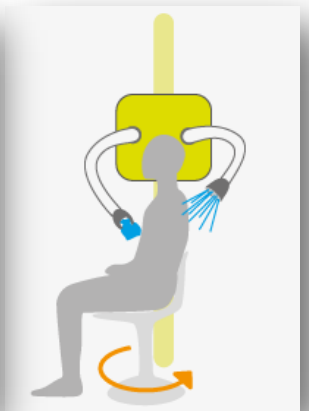
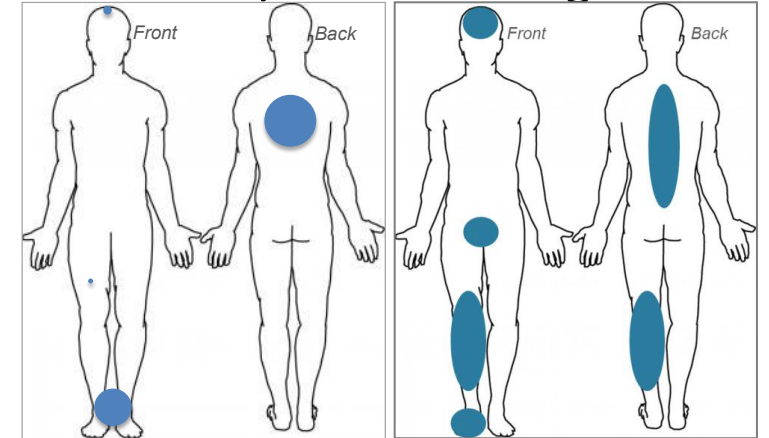
aesthetics

shower curtain fabric



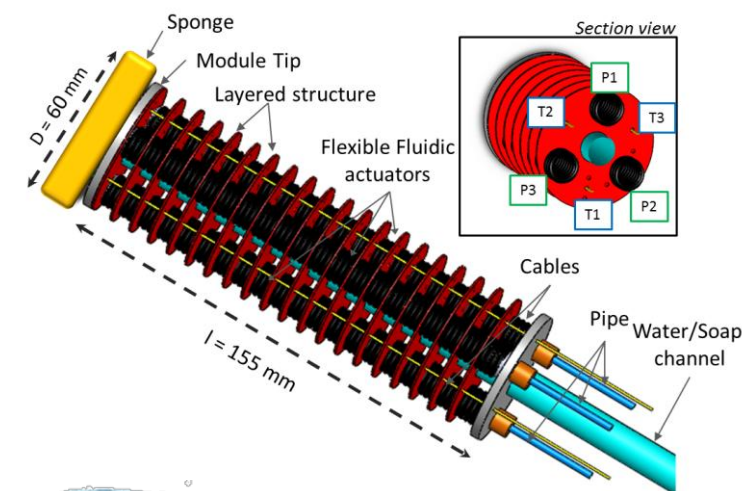
Elderly

Care givers



柔性机械臂

I-Support Robot Arm



MODULAR APPROACH

模块化方案

核心能力

1. 多平面弯曲

1. Bending on multiple planes



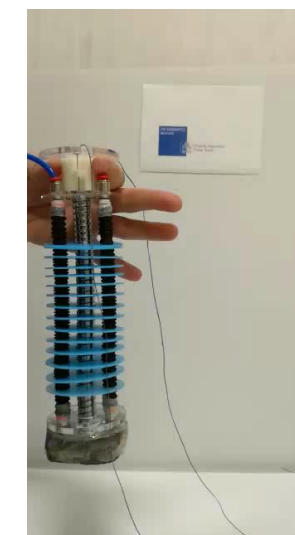
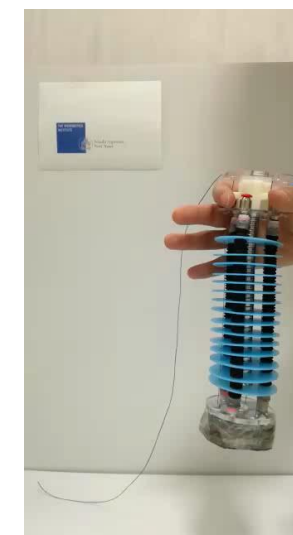
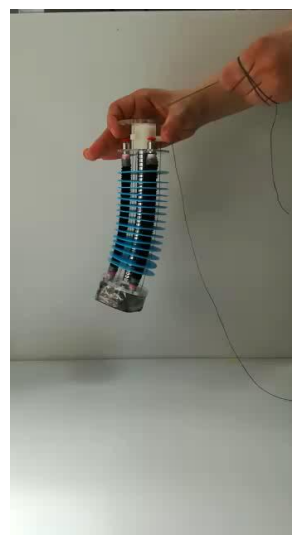
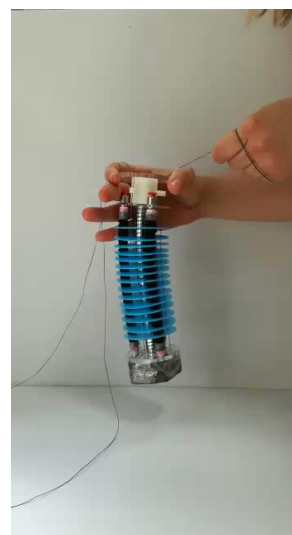
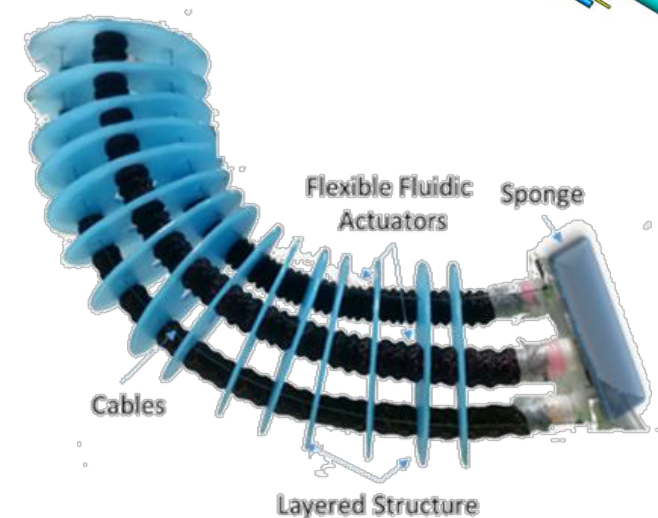
BBC World News

收缩

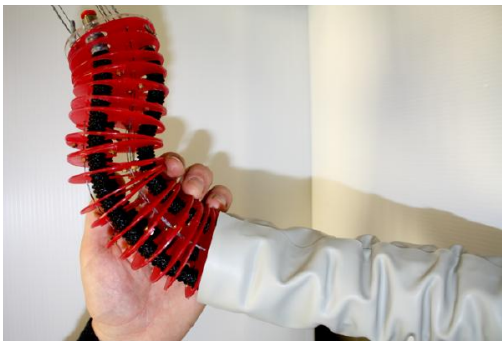
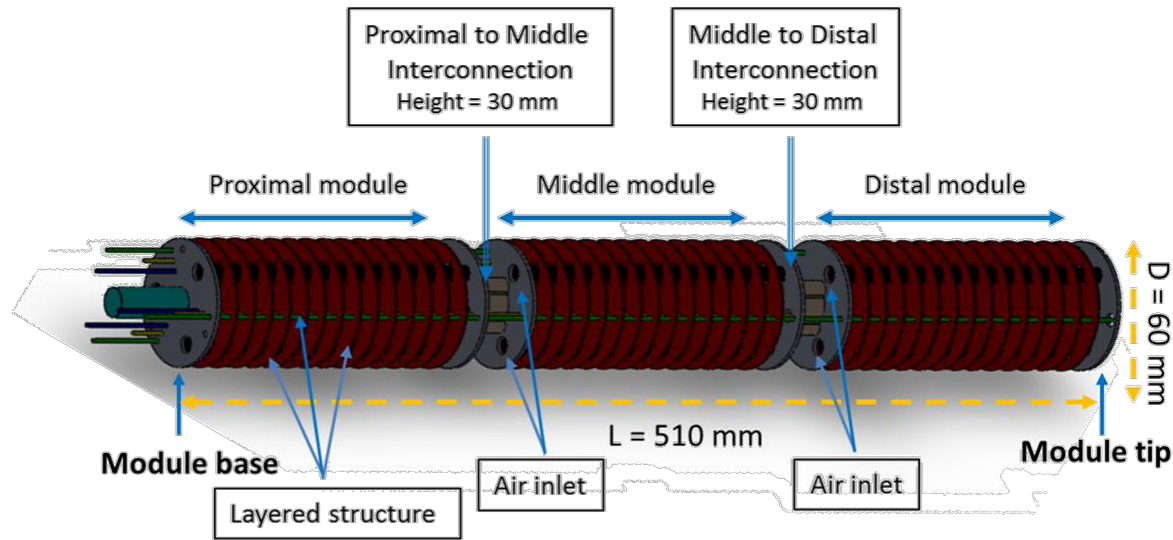
2. Contraction

刚度调节

3. Stiffness Modulation



I-Support Robot Arm



MODULAR APPROACH



CHANGABLE END-EFFECTOR



WATER-PROOF COVER

预实验（试点研究）：2个地点，分为2个阶段

Pilot Studies: 2 Phases in 2 Sites

在欧洲两处地点开展试点测试（意大利Brescia、德国Bethanien）

- Primary users（主要用户）：目标使用者（需要洗浴辅助的残障人群）
- Secondary users（次要用户）：护理人员、医疗专业人员



Pilot site 1 – Fondazione Santa Lucia (Rome, Italy)

Primary users: I-Support target user group

Secondary users: care staff, healthcare professionals

Pilot site 2 – Bethanien Hospital (Heidelberg, Germany)

Primary Users: I-Support target user group

Research team members for heuristic evaluation of usability

- TAM 技术接受模型（Technology Acceptance Model）：评估用户对系统的接受度
- Touch & Feel custom made questionnaire：定制触感体验问卷

First Pilot Study: Results for Secondary Users

Summary: The scores at TAM 8 dimensions suggest a **positive experience**.

Secondary users seem rather convinced of the **ease of use**, **output quality**, and **job relevance** of the I-Support system

Scores of “intention to use” seems promising, as it suggests the actual **feasibility** of using the system in a **real healthcare context**

Technology Acceptance Model (TAM)

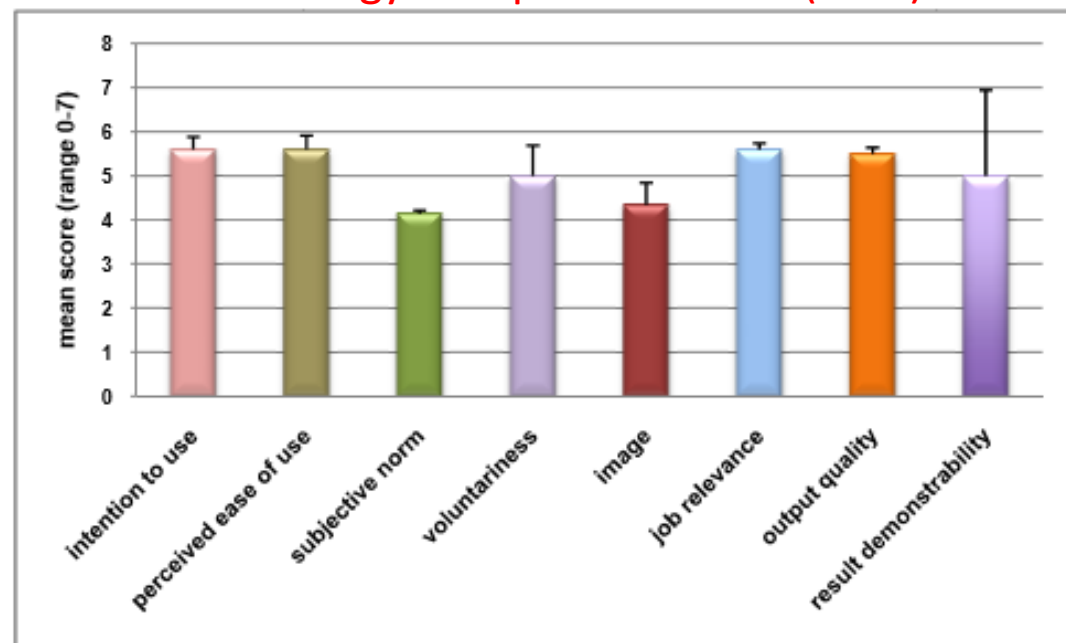


Figure 7: secondary users' feedbacks as expressed by TAM eight dimensions

Touch & Feel custom made questionnaire

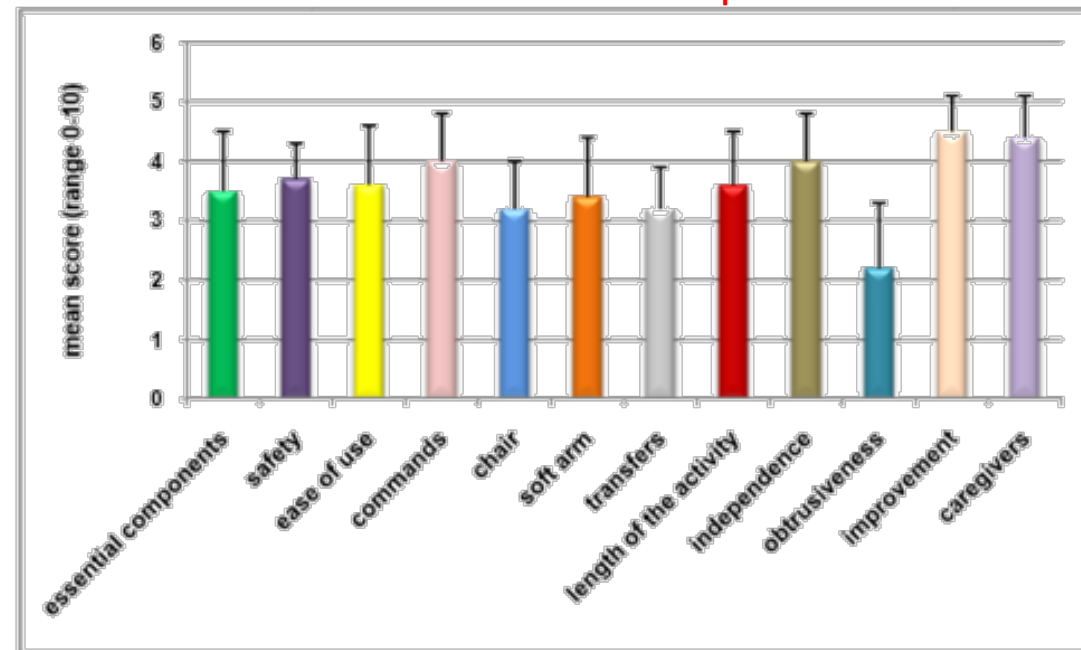


Figure 8: secondary users' feedbacks as expressed by the Touch & Feel questionnaire

Note: as a reverse item, "obtrusiveness" is coded in the opposite direction, thus it must be reverse coded (i.e., reported score of 2.2 is equivalent to 2.8)

结论摘要

TAM与触感问卷得分表明护理人员拥有积极的使用体验；认可I-Support系统在医疗场景下的实用性；使用意愿得分高，证明这套机器人系统具备落地可行性。

Second Pilot Study:

Results from primary users (@ Bethanien)

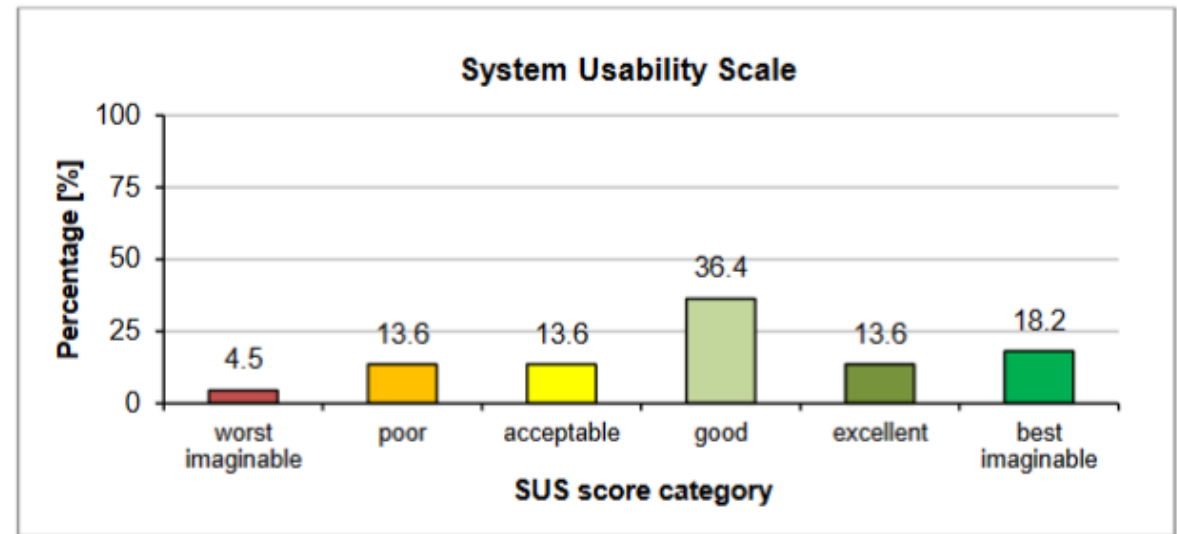
第二轮预实验：Bethanien站点，主要用户（残障使用者）结果
测试场景：向背部区域倒水（Water pouring on the back region）

SCENARIO: Water pouring on the back region

- Within- and between-subject study
- 3 different operation modes: 实验设置：
 - autonomous operation 1. 被试内研究
 - shared control 2. 3种不同操作模式
 - tele-manipulation 3. 自主操作 / 共享控制
- Evaluation of: 4. 评估指标：上背部区域的任务执行效果
 - effectiveness in showering the upper back region
 - user satisfaction in the study sample of primary users with and without cognitive impairment

结论摘要

- 自主模式（Autonomous operation）被认为是效果最好的模式
- 相比自主模式，用户控制模式的任务有效性显著降低
- 若用户存在认知障碍，需要更长时间、更多训练才能使用这套系统



Summary:

Autonomous operation seemed to be not only the **most effective**, but also the **most preferred** operation mode in the primary users of the I-SUPPORT system.

A substantially **lower effectiveness with user-controlled operation** modes has been observed, which might, however, be increased with a prolonged use or a more extensive training prior to usage.

Second Pilot Study: Results from Primary Users (@ Santa Lucia)

SCENARIO: Water pouring on the leg region

- Within- and between-subject study
- 3 different operation modes:
 - autonomous operation
 - shared control
 - tele-manipulation
- Evaluation of:
 - effectiveness in showering the leg region
 - user satisfaction in the study sample of primary users with and without cognitive impairment

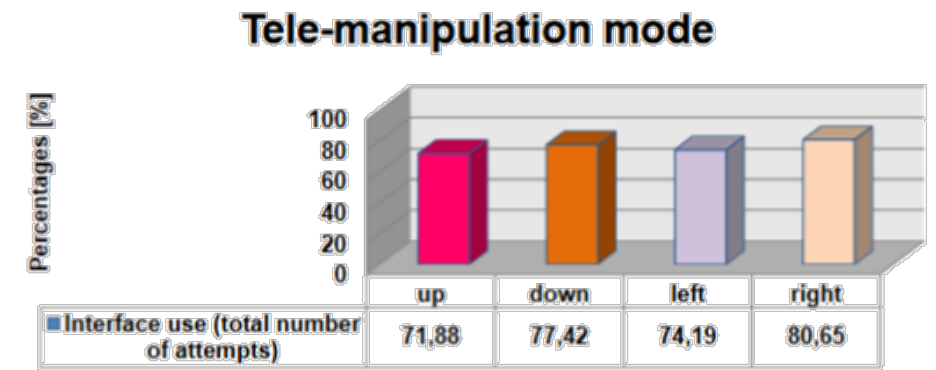


Figure 7: correct use of the tablet interface rates for tele-manipulation mode

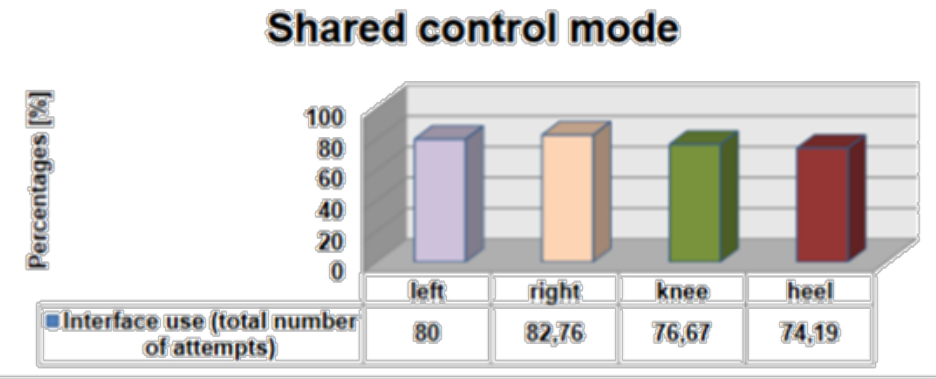


Figure 6: correct use of the tablet interface rates for shared control mode

Summary:

In both **Shared control** and **Tele-manipulation** mode, the rate of correctness of the user-interface interaction was not optimal, but promising.

Shared control: range of 74%-83% (mean percentage 78.4%),

Tele-manipulation: range of 72%-81% (mean percentage 76.03%).

Some issues related to the user-interface interaction through touchscreen

Second Pilot Study: Results from Secondary Users

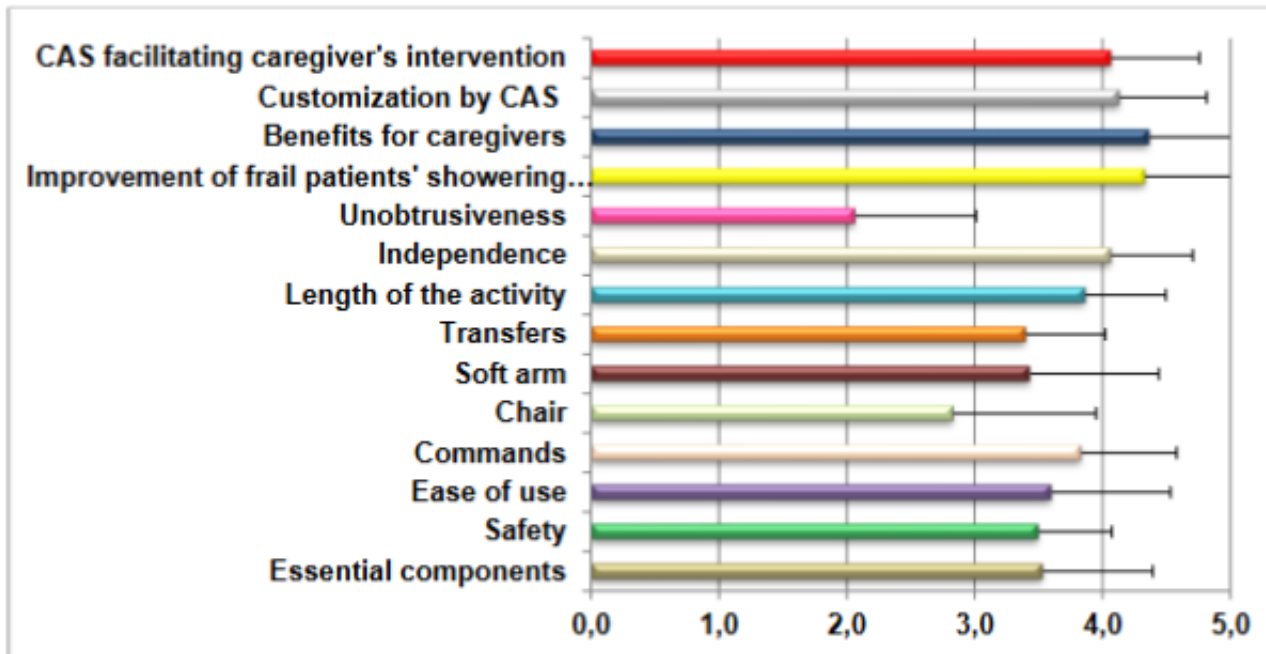


Figure 10: Secondary users' feedbacks as expressed by the Touch & Feel questionnaire

Note: as a reverse item, "unobtrusiveness" is coded in the opposite direction, thus it must be reverse coded (i.e., the reported score of 2.1 is equivalent to 2.9)

"[...] I think it could help elderly people taking shower autonomously and safely"

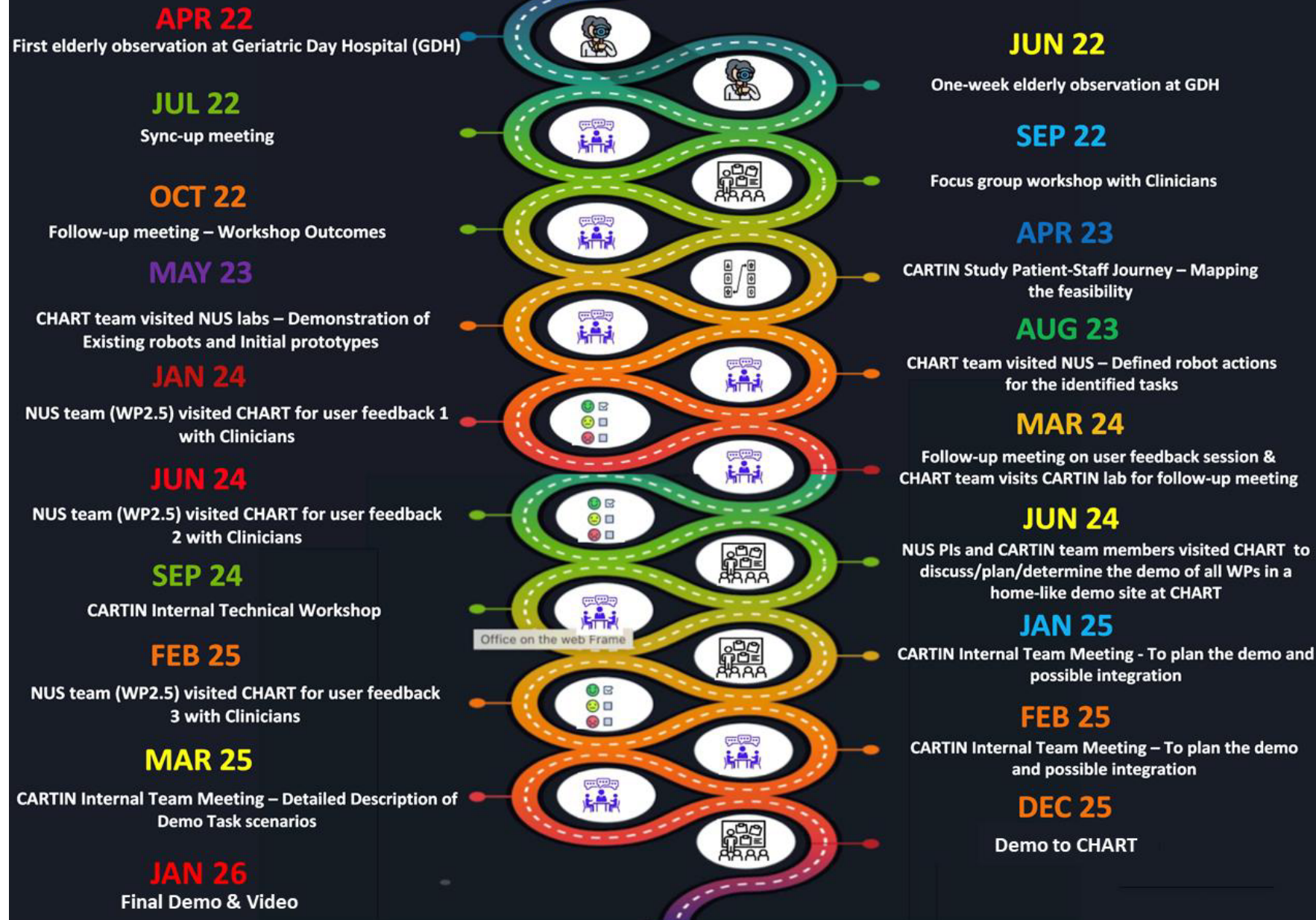
Senior Primary User

Example of Application of User-centred Design to an Assistive Robot for Eldercare

CARTIN

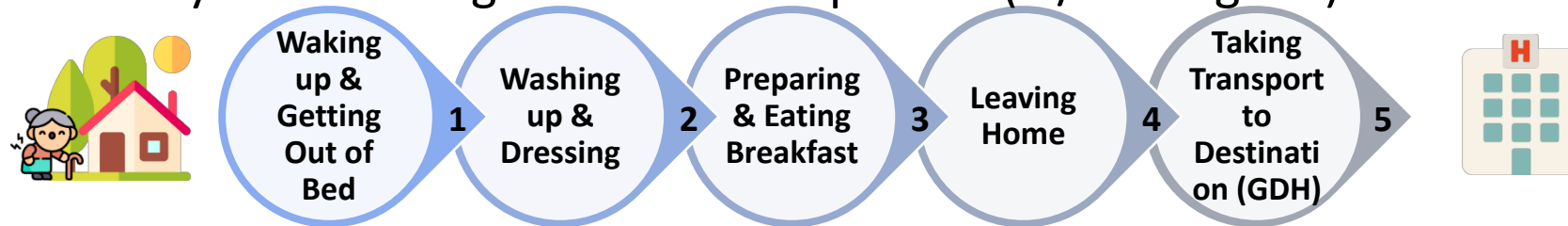


UCD (USER-CENTRED DESIGN) - COLLABORATION WITH CHART - ROADMAP



Key User Scenarios, for Patients and Caregivers

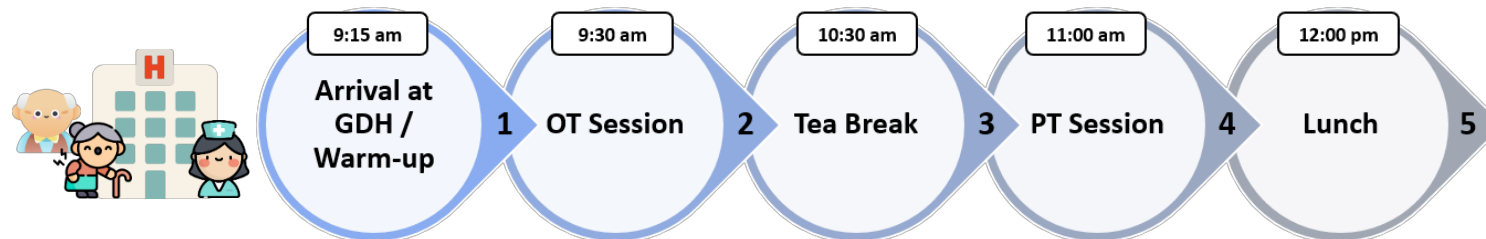
Patient's Journey – A morning in the life of a patient (w/o caregiver)



Mdm X, a 73-year old, has recently undergone a hip replacement surgery. She has been diagnosed with early dementia and goes to the Geriatric Day Hospital (GDH) once a week from 9am to 5pm for therapy to walk and to learn how to live independently as she lives at home alone.

Mobility: Some assistance required (uses mobility aids), **Cognitive:** Early stage of dementia, **Physical:** Suffers from mild hand tremors/Parkinsons, has poor strength/balance issues, **Language:** Only speaks Mother Tongue, limited understanding of English, **Environment:** Residential (alone)

Staff/Caregiver Journey – Assisting Patient at GDH



Mr Y, a 75 year old, is the caregiver for Mrs Y who is recovering from a stroke and is largely chair/bed bound. She requires help in transferring, dressing and bathing. Mr Y experiences some strain in his back due to the physically intensive tasks required to care for Mrs Y. When Mrs Y visits GDH for her therapy, Nurse A would help to assist Mrs Y, together with Mr Y.

Mobility: No issues, **Cognitive:** No issues, **Physical:** Experiences back pains due to physically intensive tasks required during caregiving, **Language:** Speaks both Mother Tongue and English, **Environment:** Residential (with caregiver)

Assistance to Transferring the Patient

- Focus on sit-to-stand, stand-to-sit and assisted walking for elderly support

Sit-to-Stand



Stand-to-Sit



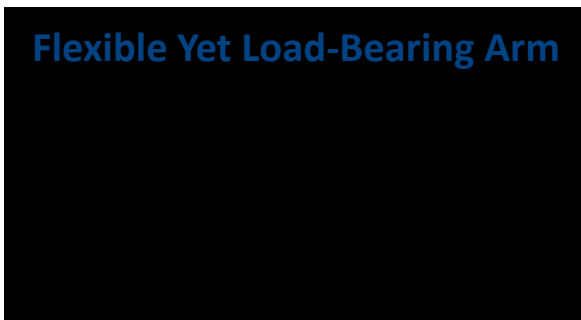
Assisted Walking



研究重点：老年人辅助场景下的坐-立转换（sit-to-stand）、立-坐转换（stand-to-sit）、辅助行走（assisted walking）

Assistive Soft Robot

Flexible Yet Load-Bearing Arm



Motion capabilities



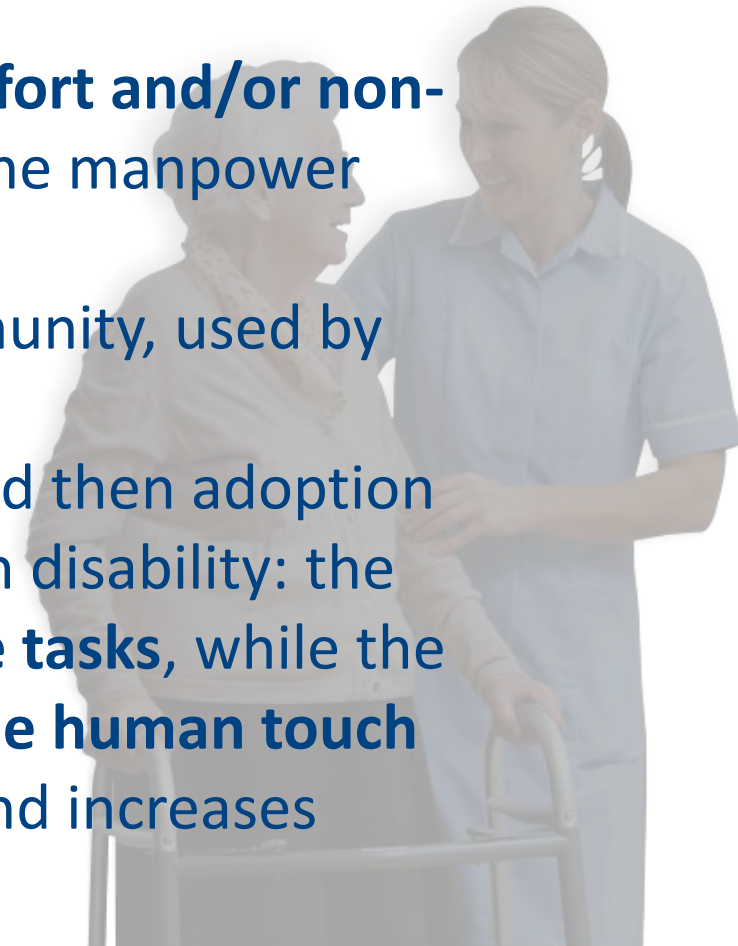
核心特性:

1. 高垂直承载能力 (High vertical load capacity) , 依靠金属骨架机构 (metallic skeleton mechanism)
2. 水平方向柔顺性 (Horizontal compliance) , 具备弯曲与伸展能力 (bending and extension capability)

- High vertical load capacity (metallic skeleton mechanism)
- Horizontal compliance with bending and extension capability

Robotics as Assistive Technology

- Many manual back-breaking tasks that require **physical effort and/or non-ergonomic postures and movements** are taking a toll to the manpower situation in healthcare
- Robots for personal assistance, either at home or in community, used by caregivers
- User-centred design approach to increase acceptability and then adoption
- Proper **allocation of roles** of caregivers and individual with disability: the robot provides **physical support and relief from repetitive tasks**, while the caregiver provides **supervision, expertise and maintain the human touch**
- Proper **human-robot role allocation** gives value to both and increases adoption opportunities for robotic assistive technologies



Robotics as Assistive Technology

- 在医疗领域，大量繁重的人工任务需要消耗大量体力，还常需要采用不符合人体工学的姿势与动作，这正持续消耗医疗人力。
- 个人辅助机器人，可用于家庭或社区场景，由护理人员使用。
- 采用以用户为中心的设计方法，提升用户接受度，进而推动技术落地应用。
- 合理分配护理人员与残障使用者的角色：机器人提供体力支持，减轻重复性工作；护理人员负责监督、提供专业判断并保持人文关怀。
- 恰当的人机角色分配对双方都有益，并能增加机器人辅助技术的落地机会。

Summary

Questions ?

- Healthcare robotics include robots used as Assistive Technology
- Assistive Technology (AT) is part of a rehabilitation process and supports individuals with disability in ADLs (Activity of Daily Life)
- AT requires design methods that include acceptability and involve users – the same for robotic assistive technologies
- Assistive robotic technologies include robotic manipulators to use as fixed workstations or wheelchair-mounted, smart wheelchairs, mobile robots, exoskeletons, robotic prostheses for upper and lower limb

Summary

Questions ?

- 医疗机器人包含用作辅助技术的机器人。
- 辅助技术（AT）是康复流程的一部分，帮助残障人群完成日常生活活动（ADLs）。
- 辅助技术需要一套包含用户接受度评估、吸纳用户参与的设计方法，机器人辅助技术亦是如此。
- 机器人辅助技术包含：固定式机器人操作臂、轮椅搭载式机械臂、智能轮椅、移动机器人、外骨骼、上下肢机器人假肢。